Looking Away

Building 1: Moves

Arrows from a floor to itself, and what counting their "hows" gives.

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A *move* is an arrow from a floor to itself that is an inside and a picture (Chapter 0, W2). Forgetting *how* one thing moves to another gives a blur: Building 0. This book keeps the how. It reads Building 0 and is never read by it (the rule F7, between buildings).

Chapter numbers restart in this book; Building 0 is cited as "Building 0, Chapter n". The rules are Chapter 0 of Building 0.

Planned, not yet written (`plans/ALGEBRA-PATH.md`, optional chapters): representations of finite groups; rings in several letters. The plane, the cubic and the torus are in Building 0 (Chapters 35–37); Part III here uses them.

Part I. Moves on things

Chapter 1. Rearrangements

a bag of n things → its moves · from: Building 0, Chapters 0, 1, 3 · build, select, blur

Select the moves of n things. Glue all swaps together and exactly two bags are left: the sign appears. Glue a move to its relabellings and the bags are the shapes of its cycles.

The first chapter that keeps the *how*. A move is an arrow from a bag to itself that is an inside and a picture (Chapter 0, W2). Building 0 forgot how one thing was carried to another; here the carrying is the thing.

Given

G1 (Building 0, Chapter 0.) Build M1–M7; blur L1–L9; arrows W1–W4; W2: inside and picture.

G2 (Building 0, Chapter 3.) $\mathbb{Z}/2$ with $+$.

G3 (Building 0, Chapter 1.) Whole numbers; counting a finite bag.

Definitions

D1 (Build, M7; Select, M4.) Fix $n \geq 1$ and $[n] := \{1, \ldots, n\}$. S_n: the maps $\sigma : [n] \to [n]$ that are one-to-one (then also onto). Each is a *move* of $[n]$. $\sigma\tau := \sigma \circ \tau$ (do τ first). e: the move that does nothing.

D2 A *swap* $(i\,j)$ exchanges i and j and keeps the rest. A *cycle* $(i_1\, i_2 \ldots i_k)$ sends each i_r to i_{r+1} and i_k to i_1.

D3 (Inversions.) $\mathrm{inv}(\sigma)$: the number of pairs $i < j$ with $\sigma(i) > \sigma(j)$.

D4 (Relabelling.) For π in S_n: $\sigma^\pi := \pi\sigma\pi^{-1}$, the move σ with the things renamed by π.

Propositions: the floor

P1 (Laws, carried by W2.) $\circ$ is associative; e does nothing on either side; every move has an undo σ^{-1}. Not commutative for $n \geq 3$: $(1\,2)(2\,3) = (1\,2\,3)$ while $(2\,3)(1\,2) = (1\,3\,2)$.

P2 (Count.) S_n has $n!$ members.

P3 (Normal form: cycles.) Every move is a product of cycles on disjoint sets of things, in exactly one way up to the order of the cycles (follow each thing until it comes back).

P4 (Swaps are enough.) Every move is a product of swaps, by P3 and

$$(i_1 \ldots i_k) = (i_1\, i_2)(i_2\, i_3) \cdots (i_{k-1}\, i_k) \quad \text{(do the right-hand swap first).}$$

Blur 1: the sign

On S_n, $n \geq 2$. *O*: $\circ$.

B1 (From below.) Gluings: $s \sim t$ for any two swaps s, t.

B2 (From above.) Question: the parity of $\mathrm{inv}(\sigma)$, in $\mathbb{Z}/2$.

P5 (Survives: parity respects $\circ$.) $\mathrm{inv}(\sigma\tau) \equiv \mathrm{inv}(\sigma) + \mathrm{inv}(\tau) \pmod 2$. (For each pair $\{i, j\}$, $\sigma\tau$ turns it over exactly when one of τ on $\{i, j\}$ and σ on $\{\tau i, \tau j\}$ does; count the pairs mod 2.) A swap has an odd number of inversions: $(i\,j)$ with $i < j$ has $2(j - i) - 1$.

P6 (Appears.) The two blurs agree. There are exactly two bags, A_n (even) and its other bag (odd). (B1 puts every product of an even number of swaps with e and every odd product with $(1\,2)$, by P4; B2 tells them apart, by P5.) Name: $\mathrm{sgn}(\sigma) := (-1)^{\mathrm{inv}(\sigma)}$.

P7 (Dies: how many swaps.) $(1\,3) = (1\,2)(2\,3)(1\,2)$: one swap, or three. The number of swaps is not a question on moves; only its parity is.

P8 (Survives into Building 0.) $\mathrm{sgn} : S_n \to \{1, -1\}$ respects $\circ$ and $\times$: an arrow onto Building 0's $\mathbb{Z}/2$ (G2), written multiplicatively. By W4, $S_n \to \mathbb{Z}/2$ is the blur B1 followed by a matching.

Blur 2: relabelling

On S_n. *O*: nothing (L7: this blur respects neither $\circ$ nor its undo in general; say so).

B3 (From below.) Gluings: $\sigma \sim \sigma^\pi$ for every π.

B4 (From above.) Question: the *cycle type*, the list of cycle lengths of P3 in falling order (a partition of n).

P9 (Appears.) The two blurs agree. $(\pi(i_1 \dots i_k)\pi^{-1} = (\pi i_1 \dots \pi i_k)$: relabelling renames the cycles and keeps their lengths; two moves of the same type are relabellings of each other.) Normal form: one move per partition of n; the bags number $p(n)$: $1, 2, 3, 5, 7, 11$ for $n = 1, \dots, 6$.

P10 (Dies: $\circ$.) $(1\,2) \sim (2\,3)$, but $(1\,2)(1\,2) = e$ while $(2\,3)(1\,2) = (1\,3\,2)$.

P11 (Survives: sign and order.) $\mathrm{sgn}(\sigma) = (-1)^{n-c}$ with c the number of cycles (counting fixed things); the order of σ (least $m \geq 1$ with $\sigma^m = e$) is the lcm of the cycle lengths. Both are read off the type.

P12 (Bag sizes.) The bag of type $1^{m_1} 2^{m_2} \cdots$ has $n! \,/\, \prod_k k^{m_k}\, m_k!$ members. (Count the relabellings that give the same move.)

Theorems

T1 (Survey: every blur of S_n that respects $\circ$.) A blur of S_n respecting $\circ$ is fixed by its bag of e, which is a union of bags of Blur 2 closed under $\circ$. For $n \leq 6$ the survey is:
$n = 2$: two blurs. $n = 3$: three (e only; A_3; everything). $n = 4$: four (e; the three double swaps with e; A_4; everything). $n = 5$ and $n = 6$: three (e; A_n; everything).

Machine-checked (all unions of types closed under $\circ$). For $n \geq 5$ the classical theorem is that A_n glues to nothing smaller: A_n is *simple*. That is a count, proved in Part II.

T2 (The sign is the only arrow to $\mathbb{Z}/2$.) For $n \geq 2$, the arrows $S_n \to \{1, -1\}$ respecting $\circ$ are $\sigma \mapsto 1$ and sgn. (Relabelling glues all swaps (P9), so an arrow takes one value on them; P4.)

T3 (Every move system lives here. Cayley.) For any finite bag G of moves of something, closed under $\circ$ and undo, $g \mapsto (h \mapsto gh)$ is an inside of G into the moves of the bag G, respecting $\circ$. So every finite move system is found inside some S_n, $n =$ its size.

Remarks

R1 (What the blur did.) The sign is an Appears line: "glue all swaps" from below meets "parity of inversions" from above. That the bags are two and not one is the kept question P5, and it is the whole content of the classical theorem.

R2 (Building 0 seen from here.) The blur of Building 0, Chapter 3 ($n \sim 0$ on $\mathbb{Z}$) is "forget how far you moved" for the moves $x \mapsto x + k$. Every blur in Building 0 is such a forgetting; this book keeps the moves.

R3 (Old names.) S_n: the symmetric group. A_n: the alternating group. P3: disjoint cycle decomposition. P6: the sign is well defined. B3: conjugacy; its bags are conjugacy classes. P12: class sizes. T1: the normal subgroups of S_n. T3: Cayley's theorem.

R4 (Machine check, `checks/b1-ch1-check.py`.) P1, P2, P4 on all of S_n, $n \leq 6$; P5 for $n \leq 5$; P6 (closure of the swap gluing); P7; P9 and P12 (types and bag sizes); P11; T1 (survey); T2.

Exercises

E1 Write $\sigma = (1\,4\,2)(3\,5)$ in S_5 as swaps, and find $\operatorname{sgn}\sigma$ and the order of σ in two ways.

E2 Count the moves of type $2\,2\,1$ in S_5 by P12, and by listing.

E3 Show that in S_n the even moves are exactly the products of 3-cycles ($n \geq 3$).

E4 Which move of S_4 has the most inversions? How many does it have?

E5 Why is "σ and τ commute" not kept by Blur 2? Give a witness in S_3.

Chapter 2. Words

letters and their undos → the free move system · from: Building 0, Chapters 0, 1, 3; Chapter 1 · build, blur

Write words in letters and their undos, and glue a letter next to its undo to nothing. Each bag holds exactly one word with nothing left to cancel. Glue more, and every move system appears.

Chapter 1 started from moves already given. This chapter builds moves from nothing but names, then blurs. It is the build-then-blur shape of Building 0, Chapter 1, one level up.

Given

G1 (Building 0, Chapter 0, M3.) Words in a list of names; maps out of words by recursion.

G2 (Chapter 1.) Moves of a bag; $\circ$; P1.

G3 (Building 0, Chapters 1, 3.) $\mathbb{Z}$ and $\mathbb{Z}/n$ with $+$.

Definitions

D1 (Build, M3.) Fix a bag L of *letters*. For each letter s take a second name s^{-1}. $W(L)$: the finite strings in the names s, s^{-1}, with the empty string 1. Joining strings, $u \cdot v$, is the operation.

D2 A word is *reduced* if no s stands next to s^{-1} (in either order).

D3 (Each letter is a move of the reduced words.) For a reduced word r: $s * r := r$ with its first name removed if that name is s^{-1}, and sr otherwise; $s^{-1} * r$ in the same way. Each of these is a move (its undo is the other), so a word $u = x_1 \cdots x_m$ acts by $u * r := x_1 * (\cdots (x_m * r))$.

The blur

On $W(L)$. O: joining.

B1 (From below.) Gluings: $s\, s^{-1} \sim 1$ and $s^{-1} s \sim 1$, for each letter s.

B2 (From above.) Question: the move $r \mapsto u * r$ of the reduced words (D3).

Propositions

P1 (Appears.) The two blurs agree. (B1 keeps the question: $s * (s^{-1} * r) = r$. Every word is glued by B1 to a reduced one, by cancelling. Two reduced words $r \neq r'$ act differently: $r * 1 = r$.)

P2 (Normal form.) Each bag holds exactly one reduced word. (P1.) The bags, with joining, form a move system $F(L)$: the *free move system* on L. Undo: $(x_1 \cdots x_m)^{-1} = x_m^{-1} \cdots x_1^{-1}$.

P3 (Count.) With two letters, the reduced words of length $k \geq 1$ number $4 \cdot 3^{k-1}$: four choices for the first name, then three (anything but the undo of the one before).

P4 (Dies: length.) $s\,s^{-1}s \sim s$, of lengths 3 and 1.

P5 (Survives: length mod 2, and each letter's exponent sum.) Cancelling removes two names, one s and one s^{-1}. So $\ell(u)$ mod 2 and $\varepsilon_s(u) :=$ (number of s) − (number of s^{-1}) survive: arrows $F(L) \to \mathbb{Z}/2$ and $F(L) \to \mathbb{Z}$, respecting joining and $+$.

P6 (One letter.) $F(\{s\}) \to \mathbb{Z}$, $u \mapsto \varepsilon_s(u)$, is a matching. Building 0, Chapter 1 builds $\mathbb{Z}$ from pairs; here it is the free move system on one letter.

Theorems

T1 (Every move system is a picture of a free one.) Let G be a move system and choose g_s in G for each letter s. There is exactly one arrow $F(L) \to G$ respecting joining with $s \mapsto g_s$. (Recursion on words, G1; it respects B1 because $g_s g_s^{-1} = e$.) If the g_s generate G, the arrow is a picture, and by W4

$$G \text{ matches } F(L) \text{ blurred by the bag of words sent to } e.$$

T2 (Presentations: glue more.) For a list R of words, glue $r \sim 1$ for each r in R, respecting joining. Write the bags $\langle L \mid R \rangle$. Then the bag of 1 is the smallest bag that holds R, joins, undos, and relabellings uru^{-1} (as Chapter 1, T1: a zero bag must be kept by relabelling).

T3 (Old floors, as presentations.)
$\langle s \mid s^n \rangle$ matches $\mathbb{Z}/n$ (Building 0, Chapter 3).
$\langle s, t \mid sts^{-1}t^{-1} \rangle$ matches $\mathbb{Z}^2$: glue $st \sim ts$ and the normal form is $s^a t^b$ (the whole-number points of $\mathbb{R}^2$, Building 0, Chapter 6).
$\langle s, t \mid s^2, t^2, (st)^3 \rangle$ matches S_3 (Chapter 1), with $s = (1\,2)$, $t = (2\,3)$.
$\langle s, t \mid s^2, t^2, (st)^n \rangle$ has $2n$ members: the moves of a regular n-gon.
$\langle s, t \mid s^2, t^3, (st)^5 \rangle$ has 60 members.

T4 (Dies: deciding.) For some finite L and R, no procedure decides whether a word lies in the bag of 1 (classical, Novikov and Boone; reported). From below, a blur can be written down and still not be computable: Building 0 never met this, because its gluings always came with a normal form.

Remarks

R1 (Build, then blur, one level up.) Building 0, Chapter 1: pairs, glue $(a+1, b+1) \sim (a, b)$, normal form $a - b$. Here: words, glue $ss^{-1} \sim 1$, normal form the reduced word. The question that makes the blur from above (D3) is itself a move: in Building 1 even the questions are moves.

R2 (Where it goes.) T1 makes every move system a blur of a free one. Part II asks which blurs of a move system respect joining (cosets, normal bags), and counts bags.

R3 (Old names.) $F(L)$: the free group on L. D3 and P1: van der Waerden's proof of the normal form. P5: abelianisation to $\mathbb{Z}^L$ (exponent sums). T1: the universal property of free groups. T2: group presentations; the bag of 1 is the normal closure of R. T3: cyclic, free abelian, symmetric, dihedral groups; the last is A_5. T4: the word problem.

R4 (Machine check, `checks/b1-ch2-check.py`.) P1 on all words of length ≤ 8 in two letters; P3 to length 8; P5; T3 (orders by coset enumeration, and the arrow onto S_3).

Exercises

E1 Reduce $s\,t\,t^{-1}s^{-1}s\,t^{-1}\,t\,s$.

E2 Show that in $F(\{s,t\})$, $st \neq ts$, using the exponent sums or P2.

E3 In $\langle s,t \mid s^2, t^2, (st)^3\rangle$, show $sts = tst$.

E4 How many reduced words of length at most 3 are there in two letters?

E5 Why is "length is even" kept by B1 but "length is ≤ 5" not?

Part II. Move systems as rows

Chapter II.1. The table of move systems

move systems → classes · from: Part I; Building 0, Chapters 1, 3, 8, 9, 14, II.1 · table, blur, class

The rows are now move systems. Every row passes the three laws of moves; the table is about what else they pass.

Building 0, Chapter II.1 again, with move systems for rows. A *move system* is a bag G with $\circ$, a do-nothing e and undos, where $\circ$ is associative: exactly the laws Chapter 1, P1 found for the moves of a bag. By Chapter 1, T3 every finite one is a bag of moves.

Given

G1 (Rows.) $\mathbb{Z}$ with $+$ (Building 0, Ch. 1); $\mathbb{Z}/6$ (Ch. 3); the circle $\mathbb{T}$ (Ch. 8); $\mathbb{Q}^\times$, the non-zero rationals with $\times$ (Ch. 2); $(\mathbb{Z}/8)^\times = \{1,3,5,7\}$ with $\times$ (Ch. 3); S_3 (Chapter 1); D_4, the moves of S_4 that carry a square's corners $1,2,3,4$ (in order round the square) to corners of the square; $GL_2(\mathbb{R})$, the straight maps of $\mathbb{R}^2$ with $\det \neq 0$ (Building 0, Ch. 14); F_2 (Chapter 2); $\mathbb{Z}^2$ (Chapter 2, T3).

G2 (Facts.) The lines of those chapters, read as yes/no facts.

Tests

D1 comm: $gh = hg$ for all g, h.

D2 fin: finitely many members.

D3 cyc: some g has every member as a power g^k, k in $\mathbb{Z}$.

D4 tf: no $g \neq e$ has $g^k = e$ for any $k \geq 1$.

D5 div: for every g and every $n \geq 1$ there is h with $h^n = g$.

The table

	comm	fin	cyc	tf	div
$\mathbb{Z}$	✓	·	✓	✓	·
$\mathbb{Z}/6$	✓	✓	✓	·	·
$\mathbb{T}$	✓	·	·	·	✓
$\mathbb{Q}^\times$	✓	·	·	·	·
$(\mathbb{Z}/8)^\times$	✓	✓	·	·	·
S_3	·	✓	·	·	·
D_4	·	✓	·	·	·
$GL_2(\mathbb{R})$	·	·	·	·	·
F_2	·	·	·	✓	·
$\mathbb{Z}^2$	✓	·	·	✓	·

Reasons for the less obvious entries: $\mathbb{T}$ is not cyc (if a turns a rational amount its powers are finitely many; if not, $a/2$ is no power of a) and is div (Building 0, Chapter 8, T2). $\mathbb{Q}^\times$: -1 has $(-1)^2 = 1$; 2 has no square root. $(\mathbb{Z}/8)^\times$: every member squares to 1. $GL_2(\mathbb{R})$: $-I$ squares to I; $\mathrm{diag}(-1, 1)$ has no square root (det of a square is ≥ 0). F_2: s has no square root (exponent sums, Chapter 2, P5); no power of a reduced word $\neq 1$ is 1.

Blur, and what appears

B1 (Building 0, II.1, B1.) For a set U of tests, glue two rows with the same answers on U.

P1 (Not sharp.) With all five tests, S_3 and D_4 share a bag. A separating test is the size: 6 against 8. Every other pair of rows is told apart.

P2 (Classes.) As T runs over the sets of tests, class(T) takes 11 values. Old names: class(comm), *abelian*; class(cyc), *cyclic* ($\mathbb{Z}$, $\mathbb{Z}/6$); class(tf), *torsion-free* ($\mathbb{Z}$, F_2, $\mathbb{Z}^2$).

P3 (Read off the table.) cyc $\Rightarrow$ comm; div $\Rightarrow$ comm. Never together: fin and tf; fin and div; cyc and div; tf and div.

Theorems

For every move system with more than one member, not only the ten rows.

T1 cyc $\Rightarrow$ comm. ($g^j g^k = g^{j+k} = g^k g^j$.)

T2 fin and tf never together. (The powers of g repeat, $g^i = g^j$ with $i < j$, so $g^{j-i} = e$.)

T3 fin and div never together. (By T2's reason each g has $g^m = e$ for some $1 \leq m \leq |G|$; so with $n = |G|!$ every n-th power is e, and $g \neq e$ has no n-th root.)

T4 cyc and div never together. (Finite: T3. Infinite: it matches $\mathbb{Z}$, and $2h = 1$ has no answer there.)

T5 (Matching rows.) $\mathbb{Z}/6$ matches $\mathbb{Z}/2 \times \mathbb{Z}/3$, and $(\mathbb{Z}/8)^\times$ matches $\mathbb{Z}/2 \times \mathbb{Z}/2$; $\mathbb{Z}/6$ and $(\mathbb{Z}/8)^\times$ differ already by size, and $\mathbb{Z}/4$ and $(\mathbb{Z}/8)^\times$ by cyc.

Remarks

R1 (Accidents of a small table; Building 0, II.1, R1.) Two lines of P3 die when the right row is added. div $\Rightarrow$ comm dies on the rotations of space: every rotation has n-th roots (turn by a $1/n$ of the angle about the same axis), and two rotations about different axes need not commute. "tf and div never together" dies on $\mathbb{Q}$ with $+$. The other four survive every row: T1–T4.

R2 (A conjecture from the table, F6.) Every abelian row with fin in the table matches a product of clocks $\mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k$. This is true for every finite abelian move system; it is proved in the chapter on modules, from a normal form of matrices.

R3 (Old names.) Move system: group. comm: abelian. tf: torsion-free. div: divisible. D_4: the dihedral group of order 8. T2: finite groups are torsion groups.

R4 (Machine check, `checks/b1-p2-ch1-check.py`.) Every finite entry by brute force; the 11 classes (`b1-p2-ch1-classes.py`); P1; T5 (the two matchings checked).

Exercises

E1 List the members of D_4 as moves of the four corners, and show D_4 is not comm.

E2 Add the row $\mathbb{Q}$ with $+$. How many classes now?

E3 Find a test that separates S_3 from D_4 without counting members.

E4 Show $\mathbb{Z}^2$ is not cyc.

Chapter II.2. Blurs that respect one side

a move system → its cosets · from: Chapter II.1; Building 0, Chapters 3, II.2 · blur, count

Ask a blur to respect moving on the left only. Its bags are copies of the bag of e, all of one size. Counting them gives Lagrange, and Fermat's little theorem falls out.

Building 0, II.2 asked which blurs of a ring respect $+$ and $\times$. Here the question is asked of a move system with less: respect $g\circ$ on the left, and nothing on the right. Rule L7 (say what the blur respects) is the whole story of this chapter and the next.

Given

G1 (Chapter II.1.) Move systems; the rows.

G2 (Building 0, Chapter 3.) $\mathbb{Z}/n$; $(\mathbb{Z}/n)^\times$, the bags with an inverse for $\times$; Chapter 3, T5 there.

The blur

On a move system G. O: for each g, the move $x \mapsto gx$ (left only).

B1 (From below.) For a bag $H \subseteq G$: glue $h \sim e$ for each h in H.

B2 (From above.) For a sub move system K (closed under $\circ$ and undo, holding e): the question $x \mapsto$ the bag $xK := \{xk : k \in K\}$.

Propositions

P1 (Survey: every such blur.) A blur of G respecting left moves is fixed by its bag of e, which is a sub move system K; and $x \sim y \iff x^{-1}y \in K$. Conversely every sub move system gives one. So these blurs and the sub move systems match.

P2 (Appears.) B1 for H equals B2 for $K =$ the smallest sub move system holding H. Its bags are the xK: the *left cosets*.

P3 (Survives: moving a bag.) $x \mapsto gx$ carries the bag yK onto the bag gyK, one-to-one. (Its undo is $x \mapsto g^{-1}x$.) So all bags have the same size as K.

P4 (Dies: the right side.) In S_3 with $K = \{e, (1\,2)\}$: $e \sim (1\,2)$, but $e(1\,3) = (1\,3)$ and $(1\,2)(1\,3) = (1\,3\,2)$ are in different bags, since $(1\,3)^{-1}(1\,3\,2) = (2\,3) \notin K$.

P5 (Normal form.) None in general: no rule picks one member of each xK. For $\mathbb{Z}$ and $K = n\mathbb{Z}$ the normal forms $0, \ldots, n-1$ of Building 0, Chapter 3 are such a rule.

Theorems

T1 (Counting bags: Lagrange. A count, not a Survives line.) If G is finite, $|G| = |K| \cdot$ (number of bags). (P3: the bags are disjoint and each has $|K|$ members.) So $|K|$ divides $|G|$.

T2 (The order of a move divides the size.) For g in a finite G, the least $m \geq 1$ with $g^m = e$ divides $|G|$. ($K = \{e, g, \ldots, g^{m-1}\}$ has m members: T1.)

T3 ($g^{|G|} = e$.) From T2.

T4 (Building 0, Chapter 3, T5, recovered.) For a with $\gcd(a, n) = 1$: $a^{\varphi(n)} \equiv 1 \pmod{n}$, where $\varphi(n) = |(\mathbb{Z}/n)^\times|$. For $n = p$ prime: $a^{p-1} \equiv 1$. (T3 in $(\mathbb{Z}/n)^\times$.)

T5 (Dies: "every divisor is a size".) A_4 has 12 members and no sub move system with 6. (Machine-checked; classical.) T1 is "at most these sizes", never "these sizes occur".

T6 (Prime size.) If $|G| = p$ is prime, G matches $\mathbb{Z}/p$. (Any $g \neq e$ has order dividing p, so p: its powers are all of G.)

Remarks

R1 (What the blur did.) The cosets are the bags of the one blur that respects left moves; their equal size is a Survives line (P3); Lagrange is a count of bags (T1). The method's own split: blur keeps (P3), counting gives existence and sizes.

R2 (Respecting both sides.) If the blur also respects right moves, K must be kept by relabelling, $gKg^{-1} = K$, and the bags themselves form a move system. That is Chapter II.3.

R3 (Old names.) Left cosets; index = number of bags; T1: Lagrange's theorem; T4: Euler's theorem, Fermat's little theorem; T5: the converse of Lagrange fails (A_4).

R4 (Machine check, `checks/b1-p2-ch2-check.py`.) P1 (for every bag $K \ni e$ of S_3 and D_4: "$x^{-1}y \in K$" is a blur exactly when K is a sub move system); P3; P4; T1–T3 on all sub move systems of S_4; T4 for $n \leq 200$; T5.

Exercises

E1 List the left cosets of $K = \{e, (1\,2)\}$ in S_3, and then the right cosets Kx. Are they the same bags?

E2 Find $3^{100} \bmod 7$ with T4.

E3 Which sizes of sub move systems occur in S_4? Compare with the divisors of 24.

E4 Show that a move system of size 4 is comm. (Each $g \neq e$ has order 2 or 4.)

Chapter II.3. Blurs that respect both sides

a move system → its quotients · from: Chapters 1, 2, II.2 · blur

Ask the blur to respect moving on both sides. The bag of e must now be kept by relabelling, and the bags form a move system of their own. Every arrow is such a blur followed by an inside.

The same gluing as Chapter II.2, with one more operation in O. Building 0, Chapter 3 is the case $G = \mathbb{Z}$, where left and right are the same and every sub move system will do.

Given

G1 (Chapter II.2.) Left-respecting blurs and cosets; T1 (Lagrange).

G2 (Chapter 1.) Relabelling σ^π; the bags of relabelling (types); T1 (survey for S_n, $n \leq 6$).

G3 (Chapter 2.) Presentations; P5 (exponent sums).

The blur

On a move system G. O: $x \mapsto gx$ and $x \mapsto xg$ for each g.

B1 (From below.) For a bag H: glue $h \sim e$ for each h in H.

B2 (From above.) For a sub move system N with $gNg^{-1} = N$ for every g (*kept by relabelling*): the question $x \mapsto xN$.

Propositions

P1 (Survey.) A blur respecting both sides is a left-respecting blur (Chapter II.2, P1) whose bag of e, N, is kept by relabelling: $n \sim e$ gives $gng^{-1} \sim gg^{-1} = e$. Conversely each such N gives one. Then $xN = Nx$: left and right bags agree.

P2 (Appears.) B1 for H equals B2 for the smallest N kept by relabelling that holds H (Chapter 2, T2).

P3 (Survives: $\circ$ on bags.) $xN \cdot yN := xyN$ does not depend on the members chosen. The bags form a move system G/N, and $x \mapsto xN$ is a picture respecting $\circ$.

P4 (Dies: "kept by relabelling" does not chain.) In D_4 (corners $1, 2, 3, 4$ round a square), let s be the flip $(2\,4)$ and $V = \{e, s, (1\,3), (1\,3)(2\,4)\}$. Then $\{e, s\}$ is kept by relabelling in V (V is comm), and V is kept in D_4 (it has half the members). But $\{e, s\}$ is not kept in D_4: the quarter turn $r = (1\,2\,3\,4)$ gives $rsr^{-1} = (1\,3)$.

P5 (Half is always kept.) If N has half the members of G, it is kept by relabelling: the two left bags and the two right bags are both N and the rest.

Theorems

T1 (Every arrow is a blur, then an inside. W4.) For an arrow $f : G \to G'$ respecting $\circ$, the blur "$f(x) = f(y)$" respects both sides; its bag of e is $N = \{x : f(x) = e\}$;

and $G/N \to G'$, $xN \mapsto f(x)$, is an inside. So G/N matches the picture of f.
Example: sgn : $S_n \to \{1, -1\}$ gives S_n/A_n matching $\mathbb{Z}/2$ (Chapter 1, P8).

T2 (Blurs of a blur.) The two-sided blurs of G/N match the two-sided blurs of G that are blurrier than N's, by $M/N \leftrightarrow M$; and $(G/N)/(M/N)$ matches G/M. Blurring twice is blurring once, with the coarser bag of e.

T3 (A sub and a blur.) For a sub move system K and N kept by relabelling: KN is a sub move system, $K \cap N$ is kept by relabelling in K, and KN/N matches $K/(K \cap N)$. (T1 for $K \to G/N$.)

T4 (Appears: the blur to comm.) Glue $xy \sim yx$ for all x, y, respecting both sides. The bags form a comm move system G^{ab}, and every arrow from G to a comm move system passes through it, in exactly one way. From above: the questions are all arrows to comm systems.
S_n^{ab} matches $\mathbb{Z}/2$, by sgn ($n \geq 2$). F_2^{ab} matches $\mathbb{Z}^2$, by the exponent sums (Chapter 2, P5). D_4^{ab} matches $\mathbb{Z}/2 \times \mathbb{Z}/2$.

T5 (Simple: only two such blurs. A count.) G is *simple* if its only two-sided blurs are "glue nothing" and "glue everything". $\mathbb{Z}/p$ is simple (Chapter II.2, T1). A_5 is simple: its relabelling bags (inside A_5) have sizes $1, 15, 20, 12, 12$; a bag of e kept by relabelling is a union of these holding the 1, and its size divides 60 (Lagrange); the only such unions have 1 or 60 members.

Remarks

R1 (L7 is the whole story.) The same gluing $h \sim e$, with left moves in O, gives cosets; with both sides, quotients. Nothing else changed.

R2 (Simple pieces.) A ladder of blurs $G \to G/N_1 \to \cdots \to \{e\}$ in which each rung's blur is simple lists the *atoms* of G (the atoms ladder of the project, rung 2: the list survives, the gluing does not). A_5 is the first atom that is not a clock.

R3 (Old names.) Kept by relabelling: normal subgroup. G/N: quotient group. T1: the first isomorphism theorem. T2: the correspondence and third isomorphism theorems. T3: the second isomorphism theorem. T4: abelianisation, $G/[G, G]$. T5: simple groups; A_5 simple. R2: Jordan–Hölder.

R4 (Machine check, `checks/b1-p2-ch3-check.py`.) P1 on all bags of S_3, D_4 (and kept-by-relabelling sub systems of S_4); P3; P4; P5 on S_4; T1 for sgn and det on $GL_2(\mathbb{Z}/3)$; T4 sizes for S_3, S_4, D_4; T5 (class sizes of A_5, all unions).

Exercises

E1 List the sub move systems of S_3 kept by relabelling.

E2 Show that every sub move system of a comm system is kept by relabelling. Which line of Building 0, Chapter 3 is this?

E3 Find S_3^{ab} by gluing $(1\,2)(2\,3) \sim (2\,3)(1\,2)$ and closing up.

E4 Why can A_5 have no arrow onto $\mathbb{Z}/2$?

Chapter II.4. Moves on matrices: the Smith form

relation matrices → their normal form · from: Chapters II.1, II.3; Building 0, Chapters 18, II.2, II.5 · blur, build

Glue a matrix to every matrix reached by moving its rows and columns. The kept questions are the gcds of the minors; the normal form is a diagonal in which each entry goes into the next. Every finite comm move system is then a product of clocks.

The first of five chapters with one template: a move system acts on matrices, the blur forgets the moves, kept questions are found, and a normal form shows they are all. Here the moves are on both sides, $A \mapsto PAQ$.

Given

G1 (Building 0, Chapter II.5.) $K[x]$; division with remainder by any $d \neq 0$ when K is a field, and in $\mathbb{Z}$; det over any ring; P10 (equations are carried); P11 (inverse iff det is a unit); T2: $\mathbb{Z}$ and $K[x]$ pass the same lines.

G2 (Building 0, Chapter 18.) gcd, and "goes into" as a one-way blur.

G3 (Building 0, Chapter II.2.) Blurs of a ring by ideals; quotients R/I.

G4 (Chapter II.3.) Quotients of move systems; T1.

Definitions

D1 R: $\mathbb{Z}$, or $K[x]$ with K a field. Its units: ± 1, or the non-zero constants.

D2 An *R-module*: a bag M with $+$ (a comm move system) and scaling by members of R, with the laws of Building 0, Chapter 6, P2. R^n is one; so is every comm move system, with $R = \mathbb{Z}$ ($k \cdot m := m + \cdots + m$).

D3 (Build, then blur.) For an $m \times n$ matrix A over R: $\operatorname{coker} A := R^n$ blurred by gluing each row of A to 0, respecting $+$ and scaling. Every finite comm move system M is a $\operatorname{coker} A$: one generator x_g for each member g, one relation $x_g + x_h - x_{g+h}$ for each pair (Chapter 2, T1 and Chapter II.3, T1, read with $+$).

D4 $d_k(A)$: the gcd of all $k \times k$ minors of A (made monic, or ≥ 0); $d_0 := 1$.

The blur

On $m \times n$ matrices over R. O: none beyond the moves themselves.

B1 (From below.) Gluings: $A \sim PAQ$ for P, Q square with det a unit (G1, P11). Equivalently, by row and column moves: swap two rows; add r times one row to another; scale a row by a unit; and the same for columns.

B2 (From above.) Questions: $d_1(A), d_2(A), \ldots$.

Propositions

P1 (Survives: the d_k.) Each $k \times k$ minor of PA is an R-combination of $k \times k$ minors of A (an equation in the entries, carried to every R: G1, P10). So $d_k(A)$ goes into $d_k(PA)$, and back again by P^{-1}: d_k is kept. The same for AQ.

P2 (Survives: the module.) $\operatorname{coker}(PAQ)$ matches $\operatorname{coker} A$. (P only remixes the relations; Q renames the generators.)

P3 (Dies: the entries.) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}$ over $\mathbb{Z}$: $d_1 = 1$, $d_2 = 6$ for both.

P4 (Normal form; at least one, by Euclid.) Every A is glued to a diagonal

$$\operatorname{diag}(e_1, e_2, \dots, e_r, 0, \dots)$$

with $e_1 \mid e_2 \mid \cdots \mid e_r$, all $\neq 0$. (Move a smallest non-zero entry to the corner; divide with remainder along its row and column; if a remainder is left, it is smaller, and repeat; sizes or degrees cannot fall forever (Building 0, Chapter 1, G8). Then clear the row and column, and make the corner go into every other entry by adding a row.)

P5 (At most one.) $e_1 e_2 \cdots e_k = d_k(A)$ up to a unit, so $e_k = d_k/d_{k-1}$: fixed by the kept questions.

P6 (Appears.) B1 and B2 agree: two matrices are glued exactly when all their d_k agree. (P1 one way; P4 and P5 the other.)

P7 (Dies over $\mathbb{Z}[x]$.) The 1×2 matrix $(2 \;\; x)$ over $\mathbb{Z}[x]$ has $d_1 = 1$, but is glued to no $(e \;\; 0)$: the multiples of 2 and x together are not the multiples of one member (the constant terms of $2a + xb$ are even). Division with remainder was the line that P4 used, and $\mathbb{Z}[x]$ fails it.

Theorems

T1 (Every module given by finitely many generators and relations.) It matches

$$R^s \times R/(e_1) \times \cdots \times R/(e_r), \qquad e_1 \mid e_2 \mid \cdots \mid e_r,$$

for exactly one s and one such list of e_i that are not units (up to units). (P2, P4, P5.)

T2 (Chapter II.1, R2, proved.) Every finite comm move system matches a product of clocks $\mathbb{Z}/e_1 \times \cdots \times \mathbb{Z}/e_r$ with $e_1 \mid \cdots \mid e_r$, in one way only. ($R = \mathbb{Z}$, $s = 0$.)

T3 (Counting them.) The comm move systems with n members, up to matching, number $\prod_p \mathrm{p}(a_p)$, where $n = \prod p^{a_p}$ and $\mathrm{p}(a)$ is the number of partitions of a. For $n = 8$: 3 ($\mathbb{Z}/8$, $\mathbb{Z}/2 \times \mathbb{Z}/4$, $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$). For $n = 72$: 6. (Split each e_i by primes: Building 0, Chapter 3, T6.)

T4 (Two clocks, again.) $\mathbb{Z}/m \times \mathbb{Z}/n$ matches $\mathbb{Z}/mn$ exactly when $\gcd(m, n) = 1$: the Smith form of $\operatorname{diag}(m, n)$ is $\operatorname{diag}(\gcd, \operatorname{lcm})$.

Remarks

R1 (The template.) A move system acting on matrices; the blur "reachable by a move"; kept questions; a normal form proving they are all. Next in Building 1: equivalence over a field (rank), similarity (Jordan), forms (Sylvester), and the cycle type of Chapter 1 was the first case.

R2 (Where existence came from.) Once: the Euclid steps of P4, a construction. Uniqueness came from kept questions (P5).

R3 (Old names.) R-module; coker A: a presented module. B1: equivalence of matrices. d_k: determinantal divisors. P4: the Smith normal form; e_i: invariant factors. T1: the structure theorem for finitely generated modules over a principal ideal domain (here Euclidean). T2: the fundamental theorem of finite abelian groups. P7: $\mathbb{Z}[x]$ is not a principal ideal domain.

R4 (Machine check, `checks/b1-p2-ch4-check.py`.) P1 (d_k kept under 200 random moves); P4–P6 against a Smith form routine on random integer matrices; P3; T3 for $n \leq 200$ by listing diagonal forms; T4.

Exercises

E1 Find the Smith form of $\begin{pmatrix} 2 & 4 & 4 \\ -6 & 6 & 12 \\ 10 & -4 & -16 \end{pmatrix}$ over $\mathbb{Z}$, and say which comm move system it presents.

E2 List the comm move systems with 36 members.

E3 Over $\mathbb{Q}[x]$, find the Smith form of $\begin{pmatrix} x & 1 \\ 0 & x \end{pmatrix}$.

E4 Show $\mathbb{Z}/4 \ncong \mathbb{Z}/2 \times \mathbb{Z}/2$ by P5, and by cyc.

Chapter II.5. A move system moving a bag

points of $X \to$ their orbits · from: Chapters 1, II.2, II.3; Building 0, Chapter 0 (W4) · blur, count

Let G move the points of a bag and glue each point to where it can be moved. The arrow $g \mapsto g \cdot x$ has its own blur, the cosets of the moves that fix x; so the size of a bag is a quotient, and counting fixed points counts the bags.

Chapter II.2 let G move itself. Here it moves anything. Every count in this chapter is the arrow's own blur (W4) read with Lagrange.

Given

G1 (Chapter II.2.) Left cosets; T1 (Lagrange), T2.

G2 (Chapter II.3.) Relabelling $x \mapsto gxg^{-1}$; "kept by relabelling"; T1 (an arrow is a blur, then an inside).

G3 (Chapter 1.) The moves S_X of a finite bag; T3 (Cayley).

Definitions

D1 A *move system G moving a bag X*: an arrow $G \to S_X$ respecting $\circ$. Write $g \cdot x$. So $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$.

D2 $G_x := \{g : g \cdot x = x\}$, the moves that fix x. $X^g := \{x : g \cdot x = x\}$, the points g fixes.

The blur

On X. O: $x \mapsto g \cdot x$ for each g.

B1 (From below.) Gluing: $x \sim g \cdot x$ for every g and x. The bags are the *orbits*; write $G \cdot x$ and X/G.

B2 (From above.) Questions: every yes-or-no question q on X with $q(g \cdot x) = q(x)$ for all g.

Propositions

P1 (Appears.) B1 and B2 agree. (The question "is it in $G \cdot x$?" is kept, and separates x from every point outside its orbit.)

P2 (A sub move system.) G_x is closed under $\circ$ and undo. Along an orbit, $G_{g \cdot x} = g\, G_x\, g^{-1}$: the moves fixing two glued points are relabellings of each other.

P3 (Dies: "orbits have one size".) S_3 moving the subsets of $\{1, 2, 3\}$: orbit sizes $1, 3, 3, 1$.

Theorems

T1 (The arrow's own blur: orbit and fixers. W4.) The arrow $g \mapsto g \cdot x$ from G onto $G \cdot x$ glues g and h exactly when $h^{-1}g$ is in G_x, that is when $gG_x = hG_x$. So the left cosets of G_x match the orbit one-to-one, and for finite G

$$|G \cdot x| \cdot |G_x| = |G|.$$

(G1, T1.) Orbit sizes divide $|G|$.

T2 (Counting the bags.) For finite G and X:

$$|X/G| \;=\; \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

(Count the pairs (g, x) with $g \cdot x = x$ by g, and by x: $\sum_x |G_x| = \sum_x |G|/|G \cdot x| = |G|$ for each orbit, by T1.)

T3 (Relabelling as moving.) G moves itself by $g \cdot x := gxg^{-1}$. The orbits are the classes (for S_n: the cycle types, Chapter 1), G_x is the set of moves that commute with x, and the fixed points of every g are the *centre* $Z(G)$. So

$$|G| \;=\; |Z(G)| + \sum |G|/|G_x|,$$

one term for each class of size above 1, each term dividing $|G|$.

T4 (Moving cosets.) G moves its own left cosets G/K by $g \cdot xK := gxK$, giving an arrow $G \to S_{G/K}$ whose blur's bag of e is kept by relabelling and lies inside K. If $|G/K|$ is the smallest prime p dividing $|G|$, then K is kept by relabelling. (The picture of G in S_p has size dividing $|G|$ and $p!$, so dividing p (G1, T1; no smaller prime divides $|G|$); the bag of e inside K then has index p, so it is K.)

Propositions on examples

P4 (Necklaces.) n beads on a ring, c colours, glued by turning: $\frac{1}{n} \sum_{d|n} \varphi(d)\, c^{n/d}$ bags. (T2: a turn by j steps fixes $c^{\gcd(j,n)}$ colourings.) For $n = 6$, $c = 2$: 14.

P5 (A cube.) The 24 turns of a cube moving the colourings of its 6 faces in 3 colours: 57 bags.

P6 (Class sizes.) S_4: $24 = 1 + 3 + 6 + 6 + 8$. A_5: $60 = 1 + 12 + 12 + 15 + 20$ (Chapter II.3, T5 used it).

Remarks

R1 (What the method did.) One blur (B1), one W4 line (T1), one count (T2). Chapter II.2 is the case $X = G$, $g \cdot x = gx$, where every orbit is all of G.

R2 (Old names.) D1: a group action; orbits; G_x: stabiliser; T1: the orbit–stabiliser theorem; T2: Burnside's lemma (Cauchy–Frobenius); T3: the class equation, centralisers; T4: the permutation representation on cosets.

R3 (Machine check, `checks/b1-p2-ch5-6-check.py`.) T1 and T2 for S_4 moving pairs; P4 against brute force for $n \leq 8$, $c = 2, 3$; P5; T3 for S_4, D_4, A_4, D_6, A_5; T4 for S_3, S_4, D_5, D_6 over all subgroups of index p generated by two members.

Exercises

E1 How many ways are there to colour the corners of a square in 3 colours, up to its 8 moves?

E2 Find G_x for S_4 moving $\{1, 2, 3, 4\}$, and check T1.

E3 Find the class sizes of S_3 and of D_4, and check T3 for each.

E4 Use T4 to show that in a move system of size $2m$, m odd, a sub move system of size m is kept by relabelling.

Chapter II.6. One prime at a time

orbits → their sizes mod p · from: Chapters II.2, II.3, II.5; Building 0, Chapter 3 · count

Let a move system of size p^n move a bag. Every orbit has size 1 or a multiple of p, so the fixed points count the bag mod p. From that one line: centres, Cauchy, and all three Sylow lines.

These are Theorems, not Survives lines: each says something exists, and the existence comes from a count, not from a blur. Each is true in every finite row of the table at once.

Given

G1 (Chapter II.5.) T1 (orbit sizes divide $|G|$), T3 (moving by relabelling; the centre), T4 (moving cosets).

G2 (Chapter II.2.) T1 (Lagrange), T6 (prime size means a clock).

G3 (Building 0, Chapter 3.) Arithmetic mod p.

Theorems

T1 (Counting bags mod p.) If $|G| = p^n$ and G moves a finite X, then

$$|X^G| \;\equiv\; |X| \pmod{p},$$

where X^G is the set of points every g fixes. (Each orbit has size dividing p^n (G1): 1, or a multiple of p. The orbits of size 1 are X^G.)

T2 (A p-system has a centre.) If $|G| = p^n$, $n \geq 1$, then $Z(G) \neq \{e\}$. (T1 with G moving itself by relabelling: $|Z(G)| \equiv |G| \equiv 0$, and e is in $Z(G)$.)

T3 (Size p^2.) A move system of size p^2 is comm: $\mathbb{Z}/p^2$ or $\mathbb{Z}/p \times \mathbb{Z}/p$. (If $G/Z(G)$ were a clock $\mathbb{Z}/p$, generated by gZ, every member would be $g^i z$, and these commute; so $Z(G) = G$ by T2. Then Chapter II.4, T2.)

T4 (Cauchy.) If p divides $|G|$, some g has order p. (Let X be the lists $(g_1, \ldots, g_p)$ with $g_1 g_2 \cdots g_p = e$: $|X| = |G|^{p-1} \equiv 0$. The clock $\mathbb{Z}/p$ moves X by turning a list, which keeps the product e (it relabels it). Fixed points: $(g, \ldots, g)$ with $g^p = e$. By T1 their number is $\equiv 0$; $(e, \ldots, e)$ is one, so there are at least $p - 1$ others.)

T5 (Sylow I: at least one.) If $|G| = p^a m$ with $p \nmid m$, G has a sub move system of size p^a. (G moves the subsets of size p^a by $S \mapsto gS$. Their number $\binom{p^a m}{p^a}$ is not divisible by p (count factors of p in top and bottom term by term), so some orbit has size not divisible by p. For S in it, $|G_S| = |G|/|G \cdot S|$ is divisible by p^a; and G_S moves S without fixed points by $g \cdot s = gs$, each orbit of size $|G_S|$, so $|G_S| \leq |S| = p^a$.)

T6 (Sylow II: all alike.) Every sub move system H of size a power of p lies in some relabelling gPg^{-1} of a given P of size p^a. In particular all such P are relabellings of each other. (H moves the cosets G/P, m of them, $m \not\equiv 0$. By T1 some gP is fixed: $HgP = gP$, so $g^{-1}Hg \subseteq P$.)

T7 (Sylow III: how many.) The number n_p of sub move systems of size p^a divides m and is $\equiv 1 \pmod{p}$. (G moves them by relabelling, one orbit (T6), so $n_p = |G|/|G_P|$,

and $P \subseteq G_P$ gives $n_p \mid m$. P moves them by relabelling; a fixed Q has P and Q both inside G_Q, where Q is kept by relabelling, so by T6 in G_Q, $P = Q$. T1: $n_p \equiv 1$.)

Propositions

P1 (Size 15 is a clock.) $n_3 \mid 5$ and $n_3 \equiv 1 \pmod 3$ give $n_3 = 1$; likewise $n_5 = 1$. Both are kept by relabelling, meet only in e, and their members commute; so G matches $\mathbb{Z}/3 \times \mathbb{Z}/5 = \mathbb{Z}/15$ (Building 0, Chapter 3, T6).

P2 (Dies: "every divisor is a size", again.) A_4, size 12: Sylow gives sub move systems of sizes 4 and 3 ($n_3 = 4$), but Chapter II.2, T5 stands: none of size 6. Sylow reaches the prime powers only.

P3 (Counts.) S_4: $n_2 = 3$, $n_3 = 4$. A_5: $n_2 = 5$, $n_3 = 10$, $n_5 = 6$. S_5: $n_2 = 15$, $n_3 = 10$, $n_5 = 6$.

Remarks

R1 (What the method did.) One line, T1, used five times with five different bags: the members (T2), lists with product e (T4), subsets (T5), cosets (T6), and the P's themselves (T7). The blur of Chapter II.5 supplies the bags; the prime supplies the count.

R2 (One prime at a time.) The table of Chapter II.1 could not separate S_3 from D_4 by tests. Sylow counts are not tests (they are not laws), but they are kept by matching; they are the first numbers that sort non-comm rows.

R3 (Old names.) $|G| = p^n$: a p-group; T1: the fixed-point congruence; T2: p-groups have non-trivial centre; T4: Cauchy's theorem (McKay's proof); T5–T7: the Sylow theorems (T5 by Wielandt's count); G_P: the normaliser.

R4 (Machine check, `checks/b1-p2-ch5-6-check.py`.) T1 for D_4 moving colourings; T2 for D_4 and a non-comm system of size 27; T4–T7 for S_4, A_4, A_5, S_5, D_6, D_{10} (every element of prime-power order lies in a conjugate of P); the binomial count of T5 for $p = 2, 3, 5$; P1.

Exercises

E1 Show that every move system of size 35 is a clock.

E2 Find n_2 and n_3 for D_6 (the 12 moves of a hexagon).

E3 Show that a move system of size p^n moving a bag of size not divisible by p fixes some point.

E4 Why can T5's count not be done with subsets of size p^{a+1}?

Chapter II.7. Units moving a ring

a domain → its divisibility · from: Chapters II.4, II.5; Building 0, Chapters 18, II.2, II.5 · blur, rerun

Building 0, Chapter 18 glued -1 to 1. In any ring the units are a move system moving the ring; glue each member to its multiples by units and rerun Chapter 18. Unique factorisation dies, and comes back exactly when division has a remainder.

A rerun, in the style of Building 0, Chapter II.5: which lines of Chapter 18 survive in other rings, and what each needs.

Given

G1 (Building 0, Chapter 18.) $\mid$ as a one-way blur; B1 ($-1 \sim 1$), P8 (Appears), P9; T1–T3 (every bag above a prime; the picture to $\mathbb{Z}/p$ keeps "no zero divisors"; unique factorisation); T8 (crossing two blurs).

G2 (Building 0, Chapters II.2, II.5.) Ideals and the blurs R/I; Frac; $K[x]$; P5 (nzd carried to $K[x]$); P6 (division with remainder).

G3 (Chapter II.5.) A move system moving a bag; orbits.

G4 (Chapter II.4.) P7: $(2 \;\; x)$ over $\mathbb{Z}[x]$.

Definitions

D1 R: a commutative ring passing nzd (a *domain*). $R^\times$: its members with an inverse, a comm move system under $\times$, moving R by $u \cdot a := ua$.

D2 $a \mid b$: $b = ac$ for some c. p is *irreducible*: not 0, not a unit, and every $d \mid p$ is a unit or *up*. p is *prime*: not 0, not a unit, and $p \mid ab$ gives $p \mid a$ or $p \mid b$.

D3 R is *Euclidean* if there is a size $\sigma : R \setminus \{0\} \to \mathbb{N}$ with division with remainder: for a and $b \neq 0$, $a = qb + r$ with $r = 0$ or $\sigma(r) < \sigma(b)$.

The blur

On R. O: $\times$.

B1 (From below.) The orbits of $R^\times$ (G3): $a \sim ua$. For $\mathbb{Z}$, Chapter 18's $-1 \sim 1$.

B2 (From above.) Questions: "$d \mid a$?", one for each d.

Propositions

P1 (Survives: Chapter 18, P8 and P9.) B1 and B2 agree, and $a \sim b \iff a \mid b$ and $b \mid a$. ($a = bc$, $b = ad$ give $a(1 - dc) = 0$; for $a \neq 0$, nzd gives $dc = 1$.) $\mid$ is an order on the bags.

P2 (Survives: prime implies irreducible.) If p is prime and $p = ab$, then $p \mid a$ say, $a = pc$, $p = pcb$, so $cb = 1$ (nzd) and b is a unit.

P3 (Dies: Chapter 18, T2 and T3, in $\mathbb{Z}[\sqrt{-5}]$.) Size $\sigma(a+b\sqrt{-5}) := a^2+5b^2$ respects $\times$. No member has size 2 or 3, so 2, 3, $1 \pm \sqrt{-5}$ (sizes $4, 9, 6, 6$) are irreducible, and no two are glued (the only units are ± 1). Yet

$$2 \cdot 3 \;=\; (1+\sqrt{-5})(1-\sqrt{-5}).$$

So 2 is irreducible and not prime, and factorisation is not unique.

Theorems

T1 (Euclidean: every ideal is the multiples of one member.) In a Euclidean R, every ideal $I \neq 0$ is $(d) := dR$ for a d of smallest size in I. (For a in I, the remainder of a by d is in I and smaller, so 0; Building 0, Chapter 1, G8.) So every blur of R respecting $+$ and $\times$ is "same remainder by d".

T2 (When every ideal is (d): Chapter 18, T1–T3 come back.) If every ideal of R is some (d), then
(a) $\gcd(a,b)$ exists: $(a)+(b)=(d)$, and $d = sa+tb$;
(b) irreducible implies prime: if $p \nmid a$, $\gcd(p,a)$ is a unit, $1 = sp+ta$, so $b = spb+tab$ and $p \mid b$ when $p \mid ab$;
(c) every non-zero non-unit is a product of irreducibles, in one way up to order and units: "is", because a chain $(a_1) \subsetneq (a_2) \subsetneq \cdots$ stops (its union is an ideal (d), and d lies in some (a_n)); "only", by (b), as in Chapter 18.

T3 (Euclidean, by an estimate: $\mathbb{Z}[i]$.) $\mathbb{Z}[i] = \{a+bi\}$ inside $\mathbb{C}$, with $\sigma(z) = |z|^2$, is Euclidean. (For $a, b \neq 0$, a/b is in $\mathbb{Q}(i)$; round each coordinate to the nearest whole number to get q; then $|a/b - q|^2 \leq \frac{1}{4}+\frac{1}{4}$, so $\sigma(a-qb) \leq \frac{1}{2}\sigma(b)$.) The only estimate in the chapter.

T4 (Two squares.) A prime $p \equiv 1 \pmod 4$ is a^2+b^2. ($x = (\frac{p-1}{2})!$ has $x^2 \equiv -1$: pair each j with $-j$ in $(p-1)! \equiv -1$, the last by pairing each member of $(\mathbb{Z}/p)^\times$ with its inverse. So $p \mid (x+i)(x-i)$ but divides neither; p is not prime in $\mathbb{Z}[i]$, so not irreducible (T2(b), T3): $p = \alpha\beta$ with $\sigma(\alpha) = \sigma(\beta) = p$.) Dies for $p \equiv 3 \pmod 4$: squares are 0 or 1 mod 4.

T5 (Survives: the picture $\mathbb{Z}[x] \to (\mathbb{Z}/p)[x]$ keeps "no zero divisors". Gauss.) If p goes into every coefficient of fg, then it goes into every coefficient of f or of g. ($(\mathbb{Z}/p)[x]$ passes nzd, G2: the twin of Chapter 18, T2.) Hence: the gcd of the coefficients is kept by $\times$; a polynomial over $\mathbb{Z}$ that factors over $\mathbb{Q}$ factors over $\mathbb{Z}$ with the same degrees; and $\mathbb{Z}[x]$ has unique factorisation (T2(c) for $\mathbb{Z}$ and for $\mathbb{Q}[x]$, glued by this line).

T6 (Eisenstein, through the same picture.) If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ over $\mathbb{Z}$ has $p \mid a_i$ for all i and $p^2 \nmid a_0$, then f does not factor over $\mathbb{Q}$. (By T5 a factoring is gh over $\mathbb{Z}$; mod p, $gh = x^n$, so g and h are x^r and x^s mod p (T2(c) in $(\mathbb{Z}/p)[x]$); then p goes into $g(0)$ and $h(0)$, and $p^2 \mid a_0$.) Examples: $x^5 - 4x + 2$; $x^{p-1} + \cdots + x + 1$, after $x \mapsto x+1$.

T7 (Crossing two blurs. Chapter 18, T8.) If every ideal is some (d) and $\gcd(a,b) = 1$, then $R/(ab)$ matches $R/(a) \times R/(b)$, respecting $+$ and $\times$. ($1 = sa+tb$; the pair (x,y) comes from $x\,tb + y\,sa$.)

Propositions on the classes

P4 (Dies: "every ideal is some (d)" in $\mathbb{Z}[x]$.) The ideal of 2 and x is not one (d) (G4). Yet $\mathbb{Z}[x]$ has unique factorisation (T5). So: Euclidean $\Rightarrow$ every ideal is (d) $\Rightarrow$ unique factorisation, and the second arrow does not reverse.

P5 (Reported.) The first arrow does not reverse either: in $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ every ideal is some (d), but there is no size with division with remainder (classical).

Remarks

R1 (What the method did.) The blur is an orbit blur. Lines that are equations or "`nzd`" survived; "unique factorisation" needed a construction (division) or a picture that keeps `nzd` (T5). The one estimate is T3.

R2 (Old names.) B1: associates; D3: Euclidean domain; T1, T2: principal ideal domains; T2(c): unique factorisation domain; T3: Gaussian integers; T4: Fermat's two-squares theorem (via Wilson); T5: Gauss's lemma; T6: Eisenstein's criterion; T7: the Chinese remainder theorem.

R3 (Machine check, `checks/b1-p2-ch7-check.py`.) P3 (sizes in $\mathbb{Z}[\sqrt{-5}]$); T3 on 2000 random divisions; T4 and Wilson for $p < 2000$; T5 on 200 random products (content); T6 on three examples; T7.

Exercises

E1 Write 13, 29 and 41 as sums of two squares by T4's route.

E2 Show that 3 is prime in $\mathbb{Z}[i]$.

E3 Factor $x^4 - 1$ into irreducibles over $\mathbb{Z}$, over $\mathbb{Z}[i]$, and over $\mathbb{Z}/5$.

E4 Why does T6 fail for $x^2 + 4$ with $p = 2$?

Chapter II.8. Arrows between vector spaces

matrices over a field → their rank; pairs → $V \otimes W$ · from: Chapters II.4; Building 0, Chapters 14, II.3, II.5 · blur, build

Chapter II.4 over a field: every e_i is 1, and only one number is kept, the rank. Then a build and a blur whose Appears line is the whole definition of $V \otimes W$.

Given

G1 (Building 0, Chapter II.3, T1–T5.) dim; $\dim V/W = \dim V - \dim W$; all over any field K.

G2 (Chapter II.4.) The blur $A \sim PAQ$; d_k kept; the Smith form.

G3 (Building 0, Chapter 14.) Λ: words in arrows, glued by $v \wedge v \sim 0$.

The blur

On $m \times n$ matrices over a field K. O: none beyond the moves.

B1 (From below.) $A \sim PAQ$, P, Q with $\det \neq 0$ (Chapter II.4, B1 with $R = K$).

B2 (From above.) Question: the *rank*, the largest k with some $k \times k$ minor $\neq 0$.

Propositions

P1 (Appears; normal form.) Each bag holds exactly one $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$, with r the rank. (Chapter II.4, P4–P6 with $R = K$: every non-zero e_i is a unit, so 1. Over a field d_k is 1 or 0, so all the d_k say only the rank.)

P2 (Row rank is column rank.) A and A^{T} have the same minors, so the same rank: the number of independent rows equals the number of independent columns.

P3 (Dies: the entries; and everything but the rank.) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

Theorems

T1 (The arrow's own blur. W4.) For a linear arrow $f : V \to W$, the blur "$f(u) = f(v)$" is $\sim_{\ker f}$, and $V/\ker f$ matches the image. So

$$\dim V = \dim \ker f + \operatorname{rank} f.$$

(G1.) An arrow $V \to V$, V finite, is one-to-one exactly when it is onto.

T2 (Arrows to K: the dual.) $V^* :=$ the linear arrows $V \to K$. For finite V, $\dim V^* = \dim V$. The arrow $V \to V^{**}$, $v \mapsto (\phi \mapsto \phi(v))$, is chosen by nothing and is a matching. Dies: a matching $V \to V^*$ chosen by nothing. (Each one needs a basis; moving the basis moves the matching.)

T3 (Build, then blur: $V \otimes W$.) Build the free vector space on the pairs (v, w).
B1: glue $(v + v', w) \sim (v, w) + (v', w)$, $(v, w + w') \sim (v, w) + (v, w')$, $(cv, w) \sim c(v, w) \sim (v, cw)$, respecting $+$ and scaling. Write $v \otimes w$ for the bag of (v, w).
B2: questions: every map $\beta : V \times W \to U$ that is linear in each place, read on the free space.
Appears: B1 and B2 agree. Read aloud: the maps out of $V \times W$ linear in each place match the linear maps out of $V \otimes W$, by $\beta \mapsto (v \otimes w \mapsto \beta(v, w))$.

T4 (Normal form.) If e_i and f_j are bases, the $e_i \otimes f_j$ are a basis of $V \otimes W$: $\dim = mn$. A member is $\sum c_{ij}\, e_i \otimes f_j$, a matrix (c_{ij}); it is one $v \otimes w$ exactly when that matrix has rank ≤ 1. Dies: "every member is some $v \otimes w$": $e_1 \otimes f_1 + e_2 \otimes f_2$ has rank 2.

Remarks

R1 (Twins.) Building 0, Chapter 14 is T3 with $V = W$ and one more gluing, $v \otimes v \sim 0$. Over $\mathbb{Z}$ the same build gives $\mathbb{Z}/m \otimes \mathbb{Z}/n = \mathbb{Z}/\gcd(m, n)$ (Chapter II.4 on the matrix $\binom{m}{n}$): two clocks of coprime sizes multiply to nothing.

R2 (What the method did.) Chapter II.4 with one change of ring. T1 is W4. The definition of $\otimes$ by its "universal property" is an Appears line, not an axiom.

R3 (Old names.) Rank; T1: rank–nullity, first isomorphism theorem for vector spaces; T2: the dual space, the canonical map to the double dual; T3: the tensor product and its universal property; T4: rank-one tensors (pure tensors).

R4 (Machine check, `checks/b1-p2-ch8-10-check.py`.) P1, P2, T1 on 40 random matrices over $\mathbb{Q}$ with random moves; T4's rank-2 member; R1 for four pairs.

Exercises

E1 Find the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ and a basis of its kernel; check T1.

E2 Show that $\mathbb{C} \otimes \mathbb{C}$ over $\mathbb{R}$ has dimension 4, and over $\mathbb{C}$ dimension 1.

E3 Is $e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_1 + e_2 \otimes f_2$ a single $v \otimes w$?

Chapter II.9. One arrow, moved by one change of basis

square matrices → their similarity bags · from: Chapters II.4, II.8; Building 0, Chapters 23, II.5 · blur

Glue M to PMP^{-1}. Let x act by M and the space is a module over $K[x]$; Chapter II.4 over $K[x]$ then gives the kept questions and the normal form. The characteristic polynomial is kept but is not enough.

Given

G1 (Chapter II.4.) Modules; coker; P4–P6, T1 over $R = K[x]$.

G2 (Building 0, Chapter II.5.) det over $K[x]$; P12 (Cayley–Hamilton, an equation).

G3 (Building 0, Chapter 23.) Over $\mathbb{C}$ every polynomial splits.

The blur

On $n \times n$ matrices over a field K. O: none beyond the moves (L7: $+$ and $\times$ are not respected; $\times$ by P on one side only would break the gluing).

B1 (From below.) $M \sim PMP^{-1}$, $\det P \neq 0$.

B2 (From above.) Questions: the invariant factors $e_1 \mid \cdots \mid e_n$ of the matrix $xI - M$ over $K[x]$ (Chapter II.4, P4, made monic).

Propositions

P1 (Survives.) tr, det, $\chi_M(x) = \det(xI - M)$, the rank of $(M - \lambda I)^k$ for every λ, k: each is kept by B1. ($xI - PMP^{-1} = P(xI - M)P^{-1}$, and Chapter II.4, P1; $\chi_M = e_1 \cdots e_n$ by Chapter II.4, P5.)

P2 (Dies: χ_M is not enough.) $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and 0 both have $\chi = x^2$, and are not glued (0 is glued only to itself). Their invariant factors are $1, x^2$ and x, x.

P3 (The module.) $V = K^n$ with $p(x) \cdot v := p(M)v$ is a $K[x]$-module V_M, and it matches $\operatorname{coker}(xI - M)$. Two matrices are glued by B1 exactly when their modules match. (P is a matching $V_M \to V_N$ that commutes with x.)

Theorems

T1 (Appears.) $M \sim N \iff xI - M$ and $xI - N$ have the same invariant factors. (P3 and Chapter II.4, T1 over $K[x]$.)

T2 (Normal form over any K.) Each bag holds exactly one block matrix whose blocks are the companion matrices of the $e_i \neq 1$: for $e = x^d + c_{d-1}x^{d-1} + \cdots + c_0$, the $d \times d$ matrix with 1s below the diagonal and last column $-c_0, \ldots, -c_{d-1}$. (Chapter II.4, T1: V_M is a product of $K[x]/(e_i)$, and on $K[x]/(e)$ with basis $1, x, \ldots, x^{d-1}$, x acts by the companion matrix.)

T3 (Cayley–Hamilton and the last factor.) $e_n(M) = 0$, and e_n is the monic polynomial of least degree killing M; $e_n \mid \chi_M$, so $\chi_M(M) = 0$. (e_n kills every $K[x]/(e_i)$; nothing smaller kills $K[x]/(e_n)$.)

T4 (Over $\mathbb{C}$: Jordan.) If χ_M splits (always over $\mathbb{C}$, G3), each bag holds exactly one block matrix of blocks $\lambda I + N_k$, N_k with 1s just above the diagonal, up to the order of blocks. The number of blocks for λ of size $\geq k$ is $\operatorname{rank}(M-\lambda)^{k-1} - \operatorname{rank}(M-\lambda)^k$. So the ranks of P1 are all the kept questions. (Split each e_i into $(x-\lambda)^a$ by Chapter II.7, T7; on $K[x]/(x-\lambda)^a$ with basis $(x-\lambda)^{a-1}, \ldots, 1$, x acts by $\lambda I + N_a$.)

T5 (Select: the diagonal bags.) M is glued to a diagonal matrix exactly when e_n has no repeated root in K and splits there. Dies for $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ($e_2 = x^2$) and, over $\mathbb{R}$, for $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ($e_2 = x^2 + 1$).

Remarks

R1 (What the method did.) No new idea: let x act, and Chapter II.4 does the rest. The only trap is P2, a kept question that is not all the kept questions.

R2 (A second road, by nearness.) Over $\mathbb{C}$ the diagonal bags come near every matrix (move the diagonal of a Jordan form apart). Cayley–Hamilton is an equation (Building 0, Chapter II.5, P12) true on those, so true everywhere by nearness: the proof in Building 0.

R3 (Old names.) B1: similarity, conjugation; T2: rational canonical form (Frobenius normal form); e_n: the minimal polynomial; T3: Cayley–Hamilton; T4: Jordan normal form; T5: diagonalisable.

R4 (Machine check, `checks/b1-p2-ch8-10-check.py`.) T1 one way on 15 random similar pairs (invariant factors over $\mathbb{Q}[x]$ by a Smith routine); $\prod e_i = \chi_M$; $e_n(M) = 0$; P2; T4's rank count separating two Jordan types with one χ.

Exercises

E1 Find the invariant factors and the Jordan form of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

E2 List the similarity bags of 3×3 matrices over $\mathbb{C}$ with $\chi = (x-1)^3$.

E3 Show that M and M^{T} are always glued.

Chapter II.10. Forms, moved by both sides at once

symmetric matrices → their signature · from: Chapters II.8, II.9; Building 0, Chapters 13, 24, 28 · blur

Glue M to $P^{\mathsf{T}}MP$. Over $\mathbb{R}$ two numbers are kept, the plus count and the minus count; over $\mathbb{C}$ only the rank; over $\mathbb{Z}/p$ the rank and whether det is a square. Allow only turning moves and the eigenvalues are kept: the highest point on the sphere finds them.

Given

G1 (Chapter II.8.) Rank, kept by both-side moves.

G2 (Building 0, Chapter 24.) Highest and lowest points: at a highest point of a map on a sphere, its covector is a multiple of that of the sphere's equation. (Building 0, Chapter 13, T7: a highest point exists on a closed bounded set.)

G3 (Building 0, Chapter 28.) $\langle\,,\rangle$, right angles, Cauchy–Schwarz.

The blur

On symmetric $n \times n$ matrices over K ($2 \neq 0$), read as forms $Q(v) = v^{\mathsf{T}}Mv$.

B1 (From below.) $M \sim P^{\mathsf{T}}MP$, $\det P \neq 0$.

B2 (From above, over $\mathbb{R}$.) Questions: the rank, and $p :=$ the largest dimension of a subspace on which $Q > 0$ (except at 0).

Propositions

P1 (Completing squares: at least one diagonal.) Every M is glued to a diagonal one. (If some $Q(v) \neq 0$, put v first and clear its row and column; if $Q = 0$ on all v but $M \neq 0$, then $Q(u+w) = 2u^{\mathsf{T}}Mw \neq 0$ for some u, w; repeat on what is left.)

P2 (Survives.) The rank (G1); det up to a non-zero square (since $\det(P^{\mathsf{T}}MP)$ is $\det(P)^2 \det M$).

P3 (Dies: the diagonal entries.) $\mathrm{diag}(1,1) \sim \mathrm{diag}(2,2)$, even over $\mathbb{Q}$: $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. And $\mathrm{diag}(1,1) \not\sim \mathrm{diag}(1,3)$ over $\mathbb{Q}$: det differs by 3, not a square (P2).

Theorems

T1 (Appears over $\mathbb{R}$: Sylvester.) Each bag holds exactly one

$$\mathrm{diag}(1,\dots,1,-1,\dots,-1,0,\dots,0),$$

with p ones and q minus ones; and (p,q) is kept. (P1, then scale each entry by $1/\sqrt{|d_i|}$. Kept: if a diagonal form has p' plus entries, the span of their basis members has $Q > 0$, and any subspace with $Q > 0$ meets the span of the others ($Q \leq 0$ there) only in 0, so has dimension $\leq p'$. So $p' = p$ of B2, and $q =$ rank $-p$.)

T2 (Over $\mathbb{C}$.) Each bag holds exactly one $\mathrm{diag}(1,\dots,1,0,\dots,0)$: only the rank is kept. (-1 is a square.)

T3 (Over $\mathbb{Z}/p$, p odd.) Each bag holds exactly one of

$$\mathrm{diag}(1,\dots,1,0,\dots,0), \qquad \mathrm{diag}(1,\dots,1,c,0,\dots,0),$$

c a fixed non-square: kept are the rank and whether det (of the non-zero part) is a square. (Every ax^2+by^2 with $a,b\neq 0$ takes the value 1: count the x with ax^2 in $\{1-by^2\}$: two sets of $(p+1)/2$ members in $\mathbb{Z}/p$ must meet. So pairs of entries can be turned into $1, ab$.)

T4 (Turning moves only: the spectral line.) Over $\mathbb{R}$, with P restricted to $P^{\mathsf{T}}P=I$ (turning moves, keeping $\langle\,,\rangle$), every symmetric M is glued to a diagonal matrix, and the diagonal entries, up to order, are kept: they are the roots of χ_M. (At a highest point v of Q on the sphere $|v|=1$, G2 gives $Mv=\lambda v$. The right-angle space $v^{\perp}$ is moved into itself by M, since $\langle Mw,v\rangle=\langle w,Mv\rangle=\lambda\langle w,v\rangle$. Repeat on $v^{\perp}$. Kept, because for turning moves $P^{\mathsf{T}}=P^{-1}$, and Chapter II.9, P1.)

Remarks

R1 (Three blurs on one bag of matrices.) PAQ (Chapter II.8: rank), PMP^{-1} (Chapter II.9: invariant factors), $P^{\mathsf{T}}MP$ (this chapter: signature). T4 is the place two of them agree: for turning moves, $P^{\mathsf{T}}=P^{-1}$.

R2 (Where existence came from.) P1: a construction (completing squares). T4: a highest point (Building 0, Chapter 13, T7), the analysis floor lending its one existence tool to algebra.

R3 (Reported.) Over $\mathbb{Q}$: two forms are glued exactly when they are glued over $\mathbb{R}$ and over every $\mathbb{Q}_p$ (Hasse–Minkowski; classical; not used).

R4 (Old names.) Quadratic forms, congruence of matrices; T1: Sylvester's law of inertia, signature; T3: classification over finite fields; T4: the spectral theorem for symmetric matrices, the principal axis theorem.

R5 (Machine check, `checks/b1-p2-ch8-10-check.py`.) T1 on 50 random forms under random moves (sign counts of eigenvalues); T4 (orthogonal diagonalisation; highest value on the sphere $\leq$ largest eigenvalue on 4000 random points); T3 for 2×2 forms over $\mathbb{Z}/5,\mathbb{Z}/7,\mathbb{Z}/11$ by listing all moves.

Exercises

E1 Find (p,q) for $x^2+4xy+y^2$ and for $xy+yz+zx$.

E2 Over $\mathbb{Z}/5$, is x^2+y^2 glued to x^2+2y^2?

E3 Find the highest value of $2x^2+2xy+2y^2$ on $x^2+y^2=1$, and where it is taken.

Chapter II.11. A field inside a field

$K \to$ fields around it · from: Chapters II.7, II.8; Building 0, Chapters 7, 23, II.3, II.5 · build, ladder, select

Building 0, Chapter 7 built $\mathbb{C}$ by gluing $x^2 \sim -1$. Do it with any irreducible f over any field. Degrees multiply up a ladder, which kills the old construction problems; and a ladder of one root per rung builds a field where f splits, unique up to matching.

Given

G1 (Building 0, Chapter II.5.) $K[x]$ over a field; division with remainder; P8: every blur of $K[x]$ respecting $+$, $\times$ is "same remainder by d".

G2 (Chapter II.7.) In $K[x]$: irreducible is prime; gcd as $sa + tb$; T6 (Eisenstein).

G3 (Building 0, Chapter II.3, T5.) A field L holding K is a vector space over K.

G4 (Building 0, Chapters 7 and 23.) $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$; every polynomial over $\mathbb{C}$ has a root.

Definitions

D1 For fields $K \subseteq L$: $[L : K] := \dim_K L$, the *degree*. α in L is *algebraic* over K if $p(\alpha) = 0$ for some $p \neq 0$ in $K[x]$.

D2 f in $K[x]$ is *separable* if $\gcd(f, f') = 1$ (f' by Building 0, Chapter 7, T7).

Theorems

T1 (Degrees multiply up a ladder.) $K \subseteq L \subseteq M$: $[M : K] = [M : L]\,[L : K]$. (If a_i is a basis of L over K and b_j of M over L, the $a_i b_j$ are a basis of M over K.)

T2 (Build: a root.) For f irreducible in $K[x]$ of degree d, $K[x]/(f)$ is a field of degree d over K, and the bag of x is a root of f in it. (For $g \not\equiv 0$, $\gcd(f, g) = 1$, so $sf + tg = 1$ and t is the inverse of g; normal forms: remainders, d of them free.)

T3 (The arrow's own blur. W4.) For α algebraic over K, the arrow $K[x] \to L$, $p \mapsto p(\alpha)$, glues p and q exactly when the *minimal polynomial* m_α goes into $p - q$; m_α is the monic generator of the ideal of p with $p(\alpha) = 0$ (Chapter II.7, T1), and is irreducible. So $K(\alpha)$, the least field holding K and α, matches $K[x]/(m_\alpha)$, and $[K(\alpha) : K] = \deg m_\alpha$.

T4 (Select: the algebraic numbers.) Let $\overline{\mathbb{Q}}$ be the members of $\mathbb{C}$ algebraic over $\mathbb{Q}$. It is closed under $+$, $-$, $\times$ and inverses, and every non-constant polynomial over $\overline{\mathbb{Q}}$ has a root in $\overline{\mathbb{Q}}$. (α, β algebraic: $\mathbb{Q}(\alpha, \beta)$ has finite degree by T1, and holds $\alpha + \beta$, $\alpha\beta$; a finite-degree field has only algebraic members, since $1, \gamma, \gamma^2, \ldots$ cannot all be independent. For the root: G4 gives one in $\mathbb{C}$, and it lies in a finite-degree field over $\mathbb{Q}$ by T1.) Selected inside $\mathbb{C}$, not built: no rule of choice is needed.

T5 (Dies: constructible.) Points built from 0 and 1 by ruler and compass have coordinates in a ladder of fields, each rung of degree 1 or 2 over the last (a line meets a line or circle by solving degree ≤ 2). So a constructible number has degree 2^k over $\mathbb{Q}$ (T1).

$\sqrt[3]{2}$ has degree 3 ($x^3 - 2$, Eisenstein): the cube cannot be doubled. $\cos 20°$ has minimal polynomial $8x^3 - 6x - 1$ (no rational root): 60° cannot be cut in three.

T6 (Build: a field where f splits.) For f of degree n over K, a ladder $K = K_0 \subseteq K_1 \subseteq \cdots$ where each rung adds one root of an irreducible factor of what is left (T2) ends with a field L in which f is a product of factors $x - \alpha_i$ and which is generated by the α_i. $[L : K] \leq n!$.

T7 (At most one, by lifting through the ladder.) If L and L' are two such fields for f, a matching $\phi : K \to K$ lifts to a matching $L \to L'$. The number of lifts from $K(\alpha)$ to L' is the number of distinct roots of $\phi(m_\alpha)$ in L': at most $\deg m_\alpha$, and exactly that when m_α is separable. (A lift is fixed by where α goes (T3), and it must go to a root of $\phi(m_\alpha)$. Then the next rung.)
So if f is separable, there are exactly $[L : K]$ moves of L that fix K (multiply the counts up the ladder, T1).

T8 (Separable, and when it is automatic.) f is separable $\iff$ f has no repeated root in any field holding K. Over $\mathbb{Q}$, or any K holding $\mathbb{Q}$, every irreducible f is separable: $f' \neq 0$ has lower degree, so $\gcd(f, f') = 1$. Dies over $\mathbb{Z}/p(t)$ (fractions of $(\mathbb{Z}/p)[t]$): $x^p - t$ is irreducible and $(x^p - t)' = 0$ (reported).

Remarks

R1 (What the method did.) T2 is Chapter 7's road 2, run with any f. T3 is W4. T6 is a ladder of Builds, one root per rung; T7 is "at most one" by lifting rung by rung, and its count is the input of the Galois chapter. T4 is a Select inside a closed field, so the book never needs to build an algebraic closure.

R2 (Old names.) $[L : K]$: degree; T1: the tower law; T2: $K[x]/(f)$, Kronecker's construction; m_α: minimal polynomial; $\overline{\mathbb{Q}}$: the field of algebraic numbers; T5: impossibility of doubling the cube and trisecting the angle (Wantzel); T6, T7: existence and uniqueness of splitting fields; T8: perfect fields.

R3 (Machine check, `checks/b1-p2-ch11-12-check.py`.) Degrees of $\sqrt[3]{2}$, $\sqrt{2} + \sqrt{3}$, $\sqrt[3]{2} + \sqrt{3}$, $\sqrt[3]{2}\omega$; the splitting field of $x^3 - 2$ has degree 6; $\cos 20°$; $\cos(2\pi/17)$ has degree 8, $\cos(2\pi/7)$ degree 3; T2 over $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/5$ by listing inverses.

Exercises

E1 Find $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ and a basis.

E2 Find the inverse of $1 + \sqrt[3]{2}$ in $\mathbb{Q}(\sqrt[3]{2})$ by T2's route.

E3 Can a regular 7-gon be drawn with ruler and compass? (Use $\cos(2\pi/7)$.)

E4 How many moves of $\mathbb{Q}(\sqrt[3]{2})$ fix $\mathbb{Q}$? Why is that fewer than the degree?

Chapter II.12. Fields with p^n members

$\mathbb{Z}/p \to$ its finite fields · from: Chapters II.2, II.5, II.11; Building 0, Chapters 3, II.5 · survives, build, count

In a field where $p = 0$, raising to the p-th power respects $+$. Its fixed points after n steps are a field with p^n members, and there is only one. Its non-zero members are a clock, by a count.

Given

G1 (Chapter II.11.) T1, T6, T7 (splitting fields: exist; unique; $[L : K]$ moves when separable).

G2 (Building 0, Chapter II.5, P7.) A polynomial of degree d over a field has at most d roots.

G3 (Chapter II.2.) Lagrange; the order of a move divides the size.

G4 (Building 0, Chapter 3.) $\mathbb{Z}/p$; T5 (Fermat).

Theorems

T1 (Size.) A finite field F holds a copy of $\mathbb{Z}/p$ for exactly one prime p (the blur of the arrow $\mathbb{Z} \to F$ is $\equiv_p$, p prime by nzd), and $|F| = p^n$, $n = [F : \mathbb{Z}/p]$.

T2 (Survives: the p-th power.) In F, $(a + b)^p = a^p + b^p$ and $(ab)^p = a^p b^p$. (p goes into $\binom{p}{j}$ for $0 < j < p$.) So $\mathrm{Fr}(a) := a^p$ is an arrow $F \to F$ respecting $+$ and $\times$; it is one-to-one ($a^p = 0$ only for $a = 0$), so onto. It fixes exactly $\mathbb{Z}/p$ (G4, and G2 for at most p roots of $x^p - x$).

T3 (Build: a field with p^n members.) In the splitting field of $x^{p^n} - x$ over $\mathbb{Z}/p$ (G1), the roots are p^n distinct members (gcd with the derivative -1 is 1), and they form a field: the fixed points of Fr^n, which respects $+$ and $\times$ (T2).

T4 (At most one.) Every field F with $q = p^n$ members is a splitting field of $x^q - x$ over $\mathbb{Z}/p$, so any two match (G1, T7). (Lagrange on $F^\times$: $a^{q-1} = 1$, so every member is a root of $x^q - x$, and there are q of them.) Write $\mathbb{F}_q$.

T5 (The non-zero members are a clock. A count.) $\mathbb{F}_q^\times$ matches $\mathbb{Z}/(q-1)$. (For $d \mid q-1$, at most d members have $a^d = 1$ (G2), so at most $\varphi(d)$ have order exactly d, if any have; the orders add to $q - 1 = \sum_{d|q-1} \varphi(d)$ (Building 0, Chapter 3); so every d has exactly $\varphi(d)$, and $d = q - 1$ has $\varphi(q - 1) \geq 1$.)

T6 (The moves of $\mathbb{F}_{p^n}$.) The moves of $\mathbb{F}_{p^n}$ respecting $+$ and $\times$ are exactly $\mathrm{Fr}^0, \ldots, \mathrm{Fr}^{n-1}$: a clock $\mathbb{Z}/n$. (At most $n = [\mathbb{F}_{p^n} : \mathbb{Z}/p]$ by G1, T7; Fr^k for $k < n$ is not the identity, since $x^{p^k} - x$ has fewer than p^n roots.)

T7 (Fields inside.) $\mathbb{F}_{p^m}$ sits inside $\mathbb{F}_{p^n}$ exactly when $m \mid n$, and then it is the fixed points of Fr^m. (T1 for "only if"; $p^m - 1 \mid p^n - 1$, so $x^{p^m} - x$ goes into $x^{p^n} - x$, for "if".)

T8 (Counting irreducibles.) $x^{p^n} - x$ is the product of all monic irreducibles over $\mathbb{Z}/p$ of degree $d \mid n$. So the number I_n of monic irreducibles of degree n has $\sum_{d|n} d\, I_d = p^n$, and

$$I_n = \frac{1}{n} \sum_{d|n} \mu(d)\, p^{n/d},$$

μ undoing the sum over divisors. For $p = 2$: $2, 1, 2, 3, 6, 9, \ldots$

Remarks

R1 (What the method did.) T2 is a Survives line that holds only when $p = 0$: in $\mathbb{Q}$, $(a + b)^2 \neq a^2 + b^2$. T3 is a Build (splitting field), T4 "at most one", T5 a count. T6 is the first full Galois example: the moves are a clock, and the fields inside (T7) are its sub move systems turned upside down.

R2 (Twins.) T8 is the count of necklaces of Chapter II.5, P4, with only the necklaces that no turn fixes: a monic irreducible of degree n is a bag of n roots that Fr turns round. And Chapter III.3 now has its ladder: the floors $\mathbb{F}_{p^m}$, on which the counts N_{p^m} live.

R3 (Old names.) Fr: the Frobenius map; T3, T4: existence and uniqueness of finite fields $\mathbb{F}_q = \mathrm{GF}(q)$; T5: the multiplicative group is cyclic (primitive roots); T6: $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) \cong \mathbb{Z}/n$; T8: Gauss's formula, Möbius inversion.

R4 (Machine check, `checks/b1-p2-ch11-12-check.py`.) For $\mathbb{F}_8$, $\mathbb{F}_9$, $\mathbb{F}_{16}$, $\mathbb{F}_{25}$, $\mathbb{F}_{27}$ built from an irreducible: T2 on random pairs, Fr onto; every member a root of $x^q - x$; T5 (exactly $\varphi(d)$ members of each order d); $\mathrm{Fr}^k \neq$ identity for $k < n$; T8 by listing irreducibles over $\mathbb{Z}/2$ ($n \leq 5$) and $\mathbb{Z}/3$ ($n \leq 3$).

Exercises

E1 Build $\mathbb{F}_4$ as $(\mathbb{Z}/2)[x]/(x^2 + x + 1)$ and write its tables for $+$ and $\times$.

E2 Find a member of order 8 in $\mathbb{F}_9 = (\mathbb{Z}/3)[x]/(x^2 + 1)$.

E3 Which fields sit inside $\mathbb{F}_{2^{12}}$?

E4 Factor $x^{16} - x$ over $\mathbb{Z}/2$.

Chapter II.13. The moves of a field and the fields they fix

sub move systems $\leftrightarrow$ middle fields · from: Chapters II.2, II.3, II.8, II.11, II.12 · blur, picture, count

For a middle field M, ask each move what it does on M. For a sub move system H, glue H to e. The two blurs are the same blur: that is the whole correspondence. Two counts make it true.

Given

G1 (Chapter II.11.) T1 (degrees multiply); T7: if L is the splitting field of a separable f over K, the moves of L fixing K number exactly $[L : K]$, and a matching of middle fields lifts to L.

G2 (Chapters II.2, II.3.) Cosets σH, Lagrange; kept by relabelling; G/N; T1 of II.3 (W4).

G3 (Chapter II.8; Building 0, Chapter II.3.) More unknowns than equations: a nonzero answer.

Definitions

D1 L: the splitting field of a separable f over K. $G := \mathrm{Gal}(L/K)$, the moves of L respecting $+$ and $\times$ and fixing every member of K. By G1, $|G| = [L : K]$. G moves the roots of f, so sits inside S_n.

D2 For a sub move system H: $L^H := \{a : \sigma(a) = a \text{ for all } \sigma \in H\}$, a middle field. For a middle field $K \subseteq M \subseteq L$: $G_M :=$ the σ in G that fix every member of M.

The blur

On G. O: $\sigma \mapsto \tau\sigma$ (moving on the left).

B1 (From below, for a sub move system H.) Glue $h \sim e$ for h in H, respecting O: the bags are the cosets σH (Chapter II.2).

B2 (From above, for a middle field M.) Question: "what does σ do on M?", that is $\sigma \mapsto \sigma|_M$.

Theorems

T1 (The first count: Artin.) For a sub move system H: $[L : L^H] = |H|$. ($\leq$: if $\alpha_1, \ldots, \alpha_m$ in L with $m > |H|$, the $|H|$ equations $\sum_i \sigma(\alpha_i)\, x_i = 0$, $\sigma \in H$, have a nonzero answer in L (G3); take one with fewest nonzero x_i, one of them 1; applying any τ in H gives another answer, and the difference has fewer nonzero entries, so is 0; so every x_i is fixed by H, and the α_i are dependent over L^H. $\geq$: L is the splitting field of f over L^H, so by G1 it has exactly $[L : L^H]$ moves fixing L^H, and H is among them.)

T2 (The second count.) For a middle field M: $|G_M| = [L : M]$. (G1 over M: L is the splitting field of f over M.)

T3 (Appears: the two blurs agree.) For $M = L^H$: $\sigma|_M = \tau|_M \iff \sigma H = \tau H$. So B1 for H is B2 for L^H, and the cosets of H match the arrows $M \to L$ that lift to G. ($\sigma|_M = \tau|_M$ means $\tau^{-1}\sigma \in G_M$; and $G_{L^H} = H$: $H \subseteq G_{L^H}$, and both have size $[L : L^H]$ by T1, T2.)

T4 (The correspondence.) $H \mapsto L^H$ and $M \mapsto G_M$ are inverse matchings between the sub move systems of G and the middle fields; bigger H goes to smaller L^H; and $[M : K] = |G|/|G_M|$, the number of bags. ($G_{L^H} = H$ by T3; $L^{G_M} \supseteq M$ and $[L : L^{G_M}] = |G_M| = [L : M]$ by T1, T2, so equal.)

T5 (Survives: both sides, exactly when M is itself a splitting field.) G_M is kept by relabelling $\iff$ $\sigma(M) = M$ for every σ in G $\iff$ M is the splitting field over K of some polynomial. Then restricting to M is a picture $G \to \mathrm{Gal}(M/K)$, onto (G1, lifting), whose own blur is G_M (W4):

$$\mathrm{Gal}(M/K) \text{ matches } G/G_M.$$

($G_{\sigma(M)} = \sigma G_M \sigma^{-1}$, and T4.)

T6 (The sign on the roots.) Let $\delta = \prod_{i<j}(\alpha_i - \alpha_j)$ over the roots of f. Each σ in G sends δ to $\mathrm{sgn}(\sigma)\delta$ (Chapter 1, P5). So $G \subseteq A_n \iff \delta \in K \iff \delta^2$, the discriminant, is a square in K. (T4: δ is fixed by $G \cap A_n$.)

Propositions on examples

P1 ($x^3 - 2$ over $\mathbb{Q}$.) Roots $\sqrt[3]{2}$, $\omega\sqrt[3]{2}$, $\omega^2\sqrt[3]{2}$, $\omega = e^{2\pi i/3}$. $[L : \mathbb{Q}] = 6$, so $G = S_3$. The three sub move systems of size 2 fix the three fields $\mathbb{Q}(\omega^k\sqrt[3]{2})$, of degree 3; A_3 fixes $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, of degree 2, and only that one is kept by relabelling (T5). The discriminant -108 is not a square (T6).

P2 ($x^3 - 3x + 1$.) Discriminant $81 = 9^2$: $G = A_3$, a clock of size 3; one root gives the other two.

P3 (A circle cut in five.) $\mathbb{Q}(\zeta)$, $\zeta = e^{2\pi i/5}$: G matches $(\mathbb{Z}/5)^\times$, a clock of size 4 ($\sigma_k : \zeta \mapsto \zeta^k$). Its sub move system of size 2, $\{\sigma_1, \sigma_4\}$, fixes $\zeta + \zeta^{-1} = \frac{-1+\sqrt{5}}{2}$: the regular pentagon needs only $\sqrt{5}$.

P4 (Finite fields.) Chapter II.12, T6 and T7 are T4 for $\mathbb{F}_{p^n}$ over $\mathbb{Z}/p$: the sub move systems of $\mathbb{Z}/n$ match the divisors of n, and so do the fields inside.

P5 (Dies: "splitting field over splitting field" does not chain.) $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[4]{2})$: each step is a splitting field ($x^2 - 2$, $x^2 - \sqrt{2}$), but $\mathbb{Q}(\sqrt[4]{2})$ is not one over $\mathbb{Q}$: it is real, and $x^4 - 2$ has two non-real roots. The twin of Chapter II.3, P4.

Remarks

R1 (What the method did.) The correspondence is an Appears line (T3): "what σ does on M" and "which coset σ is in" are one blur. What makes it true is two counts (T1, T2), and both are proved. T5 is the Survives line of Chapter II.3, read through T4.

R2 (Old names.) G: the Galois group; L/K: a Galois extension; T1: Artin's theorem; T4: the fundamental theorem of Galois theory; T5: normal extensions correspond to normal subgroups; T6: the discriminant.

R3 (Machine check, `checks/b1-p2-ch13-14-check.py`.) $|G|$ and "$G \subseteq A_n$ iff the discriminant is a square" for $x^3 - 2$, $x^3 - 3x + 1$, $x^4 - 2$, $x^4 + x^3 + x^2 + x + 1$, $x^4 - 10x^2 + 1$, $x^5 - 4x + 2$ (Galois groups by a resolvent routine); P1 degrees; P3; P5.

Exercises

E1 Draw the sub move systems of G for $x^4 - 10x^2 + 1$ ($L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$) and the matching middle fields.

E2 For $x^4 - 2$, G is the 8 moves of a square. Which middle field does the sub move system of turns fix?

E3 Find the middle fields of $\mathbb{Q}(e^{2\pi i/7})$ and their degrees.

Chapter II.14. Ladders of comm rungs, and roots

move systems → ladders of blurs · from: Chapters 1, II.3, II.5, II.6, II.7, II.13; Building 0, Chapters 10, 24 · ladder, blur

Blur a move system to a comm one, then blur what is glued, and so on. If the ladder reaches a point, adding n-th roots can follow it rung by rung; if not, no formula by roots exists. A_5 blurs to nothing comm, so the quintic has no formula.

Given

G1 (Chapter II.3.) T2 (blurs of a blur), T3, T4 (the blur to comm), T5 (A_5 simple).

G2 (Chapter II.13.) T4, T5; $G \subseteq S_n$ on the roots.

G3 (Chapter II.6.) T4 (Cauchy).

G4 (Chapter 1.) P4 (swaps are enough; adjacent swaps $(i\ i{+}1)$ are enough).

Definitions

D1 (The ladder of commutators.) $G^{(0)} := G$; $G^{(k+1)} :=$ the bag of e of the blur of $G^{(k)}$ to comm (G1, T4): generated by the $xyx^{-1}y^{-1}$. G is *solvable* if some $G^{(k)} = \{e\}$.

D2 L/K is *by roots* if there is a ladder $K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r \supseteq L$ with $K_{i+1} = K_i(\beta_i)$, $\beta_i^{n_i} \in K_i$. (Here K holds $\mathbb{Q}$.)

Theorems

T1 (Equivalently: a ladder of comm rungs.) G is solvable $\iff$ there is a ladder $G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = \{e\}$, each N_{i+1} kept by relabelling in N_i, with N_i/N_{i+1} comm. Pictures and sub move systems of solvable ones are solvable. ($G^{(k)}$ lies in every such N_k, by G1, T4.)

T2 (Small S_n climb down; big ones do not.) $S_4 \supseteq A_4 \supseteq \{e, (12)(34), (13)(24), (14)(23)\} \supseteq \{e\}$: rungs $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/2 \times \mathbb{Z}/2$. So S_n is solvable for $n \leq 4$. For $n \geq 5$, A_5 sits inside S_n, is simple (G1, T5) and not comm, so its blur to comm glues nothing: $A_5^{(1)} = A_5$, and S_n is not solvable.

T3 (At most one list of simple rungs.) In any two ladders whose rungs are simple and cannot be refined, the lists of rungs N_i/N_{i+1} agree up to order and matching. (Induction on $|G|$, comparing two first rungs N, N' through $N \cap N'$ with Chapter II.3, T3.) For $\mathbb{Z}/12$: $\mathbb{Z}/2, \mathbb{Z}/2, \mathbb{Z}/3$ in every ladder.

T4 (One root, one comm rung.) If K holds the n-th roots of 1 and $\beta^n = a \in K$, then $\mathrm{Gal}(K(\beta)/K)$ is inside the clock $\mathbb{Z}/n$. ($\sigma(\beta) = \zeta_\sigma \beta$ with $\zeta_\sigma^n = 1$, and $\sigma \mapsto \zeta_\sigma$ respects $\circ$ and is one-to-one.) And adding the n-th roots of 1 to $\mathbb{Q}$ has group inside $(\mathbb{Z}/n)^\times$, comm.

T5 (By roots $\Rightarrow$ solvable.) If L, the splitting field of f over K, is by roots, then $\mathrm{Gal}(L/K)$ is solvable. (Add all the roots of 1 needed first; then each rung is T4. Enlarge the ladder to one of splitting fields (Chapter II.13, T5), read off a ladder of comm rungs of the big group (Chapter II.13, T4, T5), and pass to the picture $\mathrm{Gal}(L/K)$ (T1).)

T6 (A quintic with no formula.) $f = x^5 - 4x + 2$ is irreducible (Eisenstein, $p = 2$; Chapter II.7, T6) and has exactly three real roots (Building 0, Chapter 10: $f(-2) < 0 < f(-1)$, $f(1) < 0 < f(2)$, $f(0) > 0$; f' has two real roots, so no more). So $G \subseteq S_5$ holds:
a 5-cycle (Cauchy, G3: $5 \mid [\mathbb{Q}(\alpha) : \mathbb{Q}] \mid |G|$; an element of order 5 in S_5 is a 5-cycle);
a swap (taking complex conjugates fixes three roots and swaps two).
A swap $(a\,b)$ and a 5-cycle c give all of S_5: some power of c sends $a \mapsto b$, so renumbering, $c = (1\,2\,3\,4\,5)$ and the swap is $(1\,2)$; relabelling by powers of c gives every adjacent swap; then G4. So $G = S_5$, not solvable (T2), and by T5 the roots of f cannot be written from $\mathbb{Q}$ with $+$, $-$, $\times$, $\div$ and n-th roots.

Remarks

R1 (What the method did.) Solvable is a ladder of blurs, each to a comm floor. "No formula" is a Dies line: the ladder of commutators stops at A_5, glued by nothing. Its proof is T5, which only climbs ladders already built; the one counting line is Cauchy.

R2 (Reported.) The converse of T5 holds: a solvable group gives a formula by roots (classical; not used).

R3 (Old names.) D1: the derived series; T1: solvable groups; T3: the Jordan–Hölder theorem; T4: Kummer theory, cyclotomic extensions; T5, T6: the Abel–Ruffini theorem, Galois's criterion.

R4 (Machine check, `checks/b1-p2-ch13-14-check.py`.) S_n solvable exactly for $n \leq 4$; A_5 equals its commutator; the ladder of S_4 has sizes $24, 12, 4, 1$; $x^5 - 4x + 2$ has three real roots and group S_5; a swap and a 5-cycle generate 120 moves; T4 for $x^3 - 2$ over $\mathbb{Q}(\omega)$ (degree 3).

Exercises

E1 Write the ladder of commutators of D_4 and of S_3.

E2 Show that every move system of size p^n is solvable (Chapter II.6, T2).

E3 Find a quintic with exactly three real roots and check whether Eisenstein applies.

E4 Why does T6's argument fail for $x^5 - 1$?

Part III. A cubic and its moves

Building 0, Chapters 35–37 drew a cubic in the plane, found the functions on it, and met the same curve over $\mathbb{C}$ as a torus. This part glues its points by lines and finds a comm move system, then reads it over $\mathbb{C}$ and over $\mathbb{Z}/p$.

Chapter III.1. Points glued by lines

sums of points → points · from: Chapters 2, II.1, II.3; Building 0, Chapters 35, 36 · build, blur

Add points as letters, and glue the three points of each line to three copies of O. The bags form a move system for free, and each bag of size 0 holds exactly one point.

The usual road defines a sum of two points by a drawing and then spends pages proving it is associative. Here the sum is a blur of a free move system, so associativity is never in doubt; the work moves to "at most one point per bag".

Given

G1 (Building 0, Chapter 36.) k with $2, 3 \neq 0$; the cubic C: $y^2 = x^3 + ax + b$ with $\Delta \neq 0$, and O. P1 (the edge meets C only at O, three times); P2 (a line meets C in at most 3, counted); P3 (the mirror); T1 (Build: two points of $C(k)$ on a line give a third, and the counts add to exactly 3); P10 (no function has weight 1); P11.

G2 (Chapter 2, T1; Chapter II.1.) The free comm move system on a bag: finite sums $\sum n_P P$, n_P in $\mathbb{Z}$.

G3 (Chapter II.3.) The bags of a blur of a move system that respects both sides form a move system; for a comm one, respecting + is enough.

Definitions

D1 (Build.) $\mathbb{Z}[C]$: the free comm move system on the points of $C(k)$ (G2).
$\deg \sum n_P P := \sum n_P$.

D2 For a line L, $L \cdot C :=$ its meeting points with their counts, as a member of $\mathbb{Z}[C]$: $P + Q + R$, or $2P + R$ for a tangent, or $3O$ for the edge.

The blur

On $\mathbb{Z}[C]$. O: $+$.

B1 (From below.) Gluing: $L \cdot C \sim 3\,O$ for every line L that meets C at points of $C(k)$ (G1, T1: all three are then in $C(k)$). Respects $+$. Write $\mathrm{Pic}(C)$ for the bags.

B2 (From above.) Questions: the degree; and, for each D of degree 0, "is D glued to 0?", read through functions: $D \sim 0$ exactly when D is the list of zeros minus poles of a product of powers of lines (P2).

Propositions

P1 (Free: a move system.) $\mathrm{Pic}(C)$ is a comm move system (G3). Its law is associative because $+$ on $\mathbb{Z}[C]$ is. Nothing was drawn.

P2 (What a gluing is.) The affine line $\ell = \alpha x + \beta y + \gamma$ is a function on C (Building 0, Chapter 36) whose zeros, counted, are the affine points of $L \cdot C$, and whose only pole is at O, of order $w(\ell) = 3$ or 2 (P11 there). So $L \cdot C - 3O$ is "zeros minus poles" of ℓ, and $D \sim 0$ exactly when D is zeros minus poles of $\prod \ell_i^{n_i}$.

P3 (Survives: the degree.) Each gluing has degree $3 - 3 = 0$. So deg is kept, and $\mathrm{Pic}^0(C)$, the bags of degree 0, is a sub move system.

P4 (Two moves on points.) For P, Q in $C(k)$ let $P * Q$ be the third point on their line (G1, T1). Then

$$P + Q + (P * Q) \sim 3O, \qquad R + \bar{R} \sim 2O,$$

the second by the upright line through R and its mirror $\bar{R}$ (G1, P3). So $P + Q \sim O + \overline{P * Q}$: two points are glued to one point plus O.

Theorems

T1 (At least one: normal form.) Every D of degree 0 is glued to $P - O$ for some P in $C(k)$. (Write $D = \sum(P_i - O) - \sum(Q_j - O)$. $-(Q - O) \sim \bar{Q} - O$ by P4. Then $(P-O)+(Q-O) \sim \overline{P * Q} - O$ by P4, and repeat: each step is a Build of Building 0, Chapter 36, T1.)

T2 (At most one.) $P - O \sim Q - O$ only when $P = Q$. (Put $R := \overline{P * \bar{Q}}$; then $P - Q \sim R - O$, so $R - O$ is zeros minus poles of some $f = \prod \ell_i^{n_i}$ (P2). If $R \neq O$, f has no pole on the affine part, so f lies in $k[C]$ (classical, proof deferred: a function on C with no affine pole is a polynomial function); its only pole is a simple one at O, so $w(f) = 1$; Building 0, Chapter 36, P10 says no such f exists. So $R = O$, that is $P = Q$.)

T3 (Appears: the points are a move system.) $P \mapsto$ the bag of $P - O$ matches $C(k)$ one-to-one with $\mathrm{Pic}^0(C)$ (T1, T2). Carried back to the points, the law is

$$P \oplus Q = \overline{P * Q}, \qquad O \text{ does nothing}, \qquad \ominus P = \bar{P},$$

and $P \oplus Q \oplus R = O$ exactly when P, Q, R lie on one line. Associativity of $\oplus$ is P1, carried.

T4 (Formulas.) For $P = (x_1, y_1)$, $Q = (x_2, y_2)$ with $x_1 \neq x_2$: $P \oplus Q = (x_3, -y_3)$ with x_3, y_3 as in Building 0, Chapter 36, T1. $2P$: the same with the tangent's slope $(3x_1^2 + a)/(2y_1)$; $2P = O$ when $y_1 = 0$.

Propositions on examples

P5 (Over $\mathbb{Q}$, a clock.) $y^2 = x^3 + 1$: the points O, $(-1, 0)$, $(0, \pm 1)$, $(2, \pm 3)$, and $(2, 3)$ has order 6: $C(\mathbb{Q})$ holds a $\mathbb{Z}/6$.

P6 (Over $\mathbb{Q}$, no clock.) $y^2 = x^3 - 2$, $P = (3, 5)$: $2P = (\frac{129}{100}, -\frac{383}{1000})$. The denominators grow with each doubling, and P has no finite order.

P7 (Dies when $\Delta = 0$.) On $y^2 = x^3$ T2 fails at the bad point, which every construction through it returns. Leave that point out and the rest is again a move system, matching $(k, +)$ by $(x, y) \mapsto x/y$ (machine-checked over $\mathbb{Z}/p$; classical).

Remarks

R1 (What the method did.) Associativity, the hard line of the usual road, is free (P1): the blur is of a free move system. Existence of a normal form is Euclid-like steps (T1), each a Build. The one "at most one" line, T2, is Building 0, Chapter 36, P10, plus one classical step, deferred. A second road to T2 over $\mathbb{C}$ is Chapter III.2.

R2 (Kept questions.) The two questions of B2 are all: degree, and "zeros minus poles of a function". The same template as Chapter II.4: free thing, gluing, kept questions, normal form.

R3 (Old names.) $\mathbb{Z}[C]$: divisors; B1: linear equivalence (lines generate it on a cubic); $\mathrm{Pic}^0(C)$: the degree-zero Picard group; T3: the group law on an elliptic curve; T2: "no function with a single simple pole", a case of Riemann–Roch; P4: the chord-and-tangent construction; P6: a point of infinite order; T3 with $k = \mathbb{Q}$ and finitely many generators: Mordell–Weil (classical, reported, not used).

R4 (Machine check, `checks/b1-p3-check.py`.) Over $\mathbb{Z}/p$, $p = 5, 7, 11, 13$, every smooth (a, b): O does nothing, mirror undoes, $\oplus$ commutes, associativity on all triples (on 12 points per curve when there are more); P5; P6.

Exercises

E1 On $y^2 = x^3 + 1$ over $\mathbb{Q}$, compute $2(2, 3)$ and $3(2, 3)$.

E2 Show that $P \oplus P = O$ exactly when $P = O$ or $y(P) = 0$. How many such P are there over $\mathbb{C}$?

E3 Why does B1 include the edge line, and what does it glue?

E4 Over $\mathbb{Z}/5$, list $C(\mathbb{Z}/5)$ for $y^2 = x^3 + 1$ and find which clock it is.

Chapter III.2. The torus carries the law

$\mathbb{C}/\Lambda \to C(\mathbb{C})$ · from: Chapter III.1; Building 0, Chapters 8, 37 · arrow

Three points of the torus add to 0 exactly when their images lie on one line. So the arrow of Building 0, Chapter 37 respects the moves, and + on the torus is the chord law.

Given

G1 (Building 0, Chapter 37.) Λ, $\mathbb{C}/\Lambda$ with +; $\wp$; T4 (zeros and poles equal in number; their sums differ by a member of Λ); T8 (the arrow $\phi : z \mapsto (\wp(z) : \wp'(z) : 1)$ matches $\mathbb{C}/\Lambda$ with $C(\mathbb{C})$, C: $y^2 = 4x^3 - g_2x - g_3$).

G2 (Chapter III.1.) $\oplus$; T3: $P \oplus Q \oplus R = O$ exactly when P, Q, R lie on one line. (The 4 and the g_i change nothing: scale y by 2.)

Theorems

T1 (Lines are sums in Λ.) $\phi(z_1), \phi(z_2), \phi(z_3)$ lie on one line, with counts, exactly when $z_1 + z_2 + z_3 \in \Lambda$. (A line $\alpha x + \beta y + \gamma = 0$ with $\beta \neq 0$ gives the elliptic map $\alpha\wp + \beta\wp' + \gamma$, with one triple pole per bag, at 0. Its zeros are the z whose images lie on the line (G1, T8): three of them, and their sum is in Λ (G1, T4). For $\beta = 0$: $\wp - c$ has zeros z, $-z$, and the third meeting point is $O = \phi(0)$; again the sum is 0. Conversely, the line through $\phi(z_1)$ and $\phi(z_2)$ meets C a third time at some $\phi(z_3')$, and $z_3' \sim -z_1 - z_2$.)

T2 (Survives: the arrow respects the moves.) $\phi(z_1 + z_2) = \phi(z_1) \oplus \phi(z_2)$. (T1 and G2: both sides are the point R with $\phi(z_1) \oplus \phi(z_2) \oplus \ominus R = O$, and $\ominus\phi(z) = \phi(-z)$, the mirror, since $\wp$ is even and $\wp'$ odd.)

T3 (A second road to Chapter III.1, T2, over $\mathbb{C}$.) For a cubic reached by T8, $P \mapsto P - O$ is one-to-one: the torus side has no gluing to undo. And associativity of $\oplus$ is the associativity of + on $\mathbb{C}$, carried.

T4 (Points of finite order.) The P with $nP = O$ in $C(\mathbb{C})$ match $(\frac{1}{n}\Lambda)/\Lambda$, so there are n^2 of them and they form $\mathbb{Z}/n \times \mathbb{Z}/n$. For $n = 2$: O and the three points with $y = 0$. For $n = 3$: the nine points where the tangent meets C three times.

Remarks

R1 (What the method did.) One arrow. Its respecting + is one line of Building 0, Chapter 37 (T4, the sum of zeros) read through Chapter III.1, T3.

R2 (Reported.) Every smooth cubic over $\mathbb{C}$ is reached by some Λ (classical, uniformisation). With it, T3 covers every cubic over a k inside $\mathbb{C}$, and the deferred step of Chapter III.1, T2 is no longer needed there. Not used.

R3 (Old names.) T1: Abel's theorem on the torus; T2: the addition theorem of $\wp$; T4: $E[n] \cong (\mathbb{Z}/n)^2$; the $n = 3$ points: flexes.

R4 (Machine check, `checks/b1-p3-check.py`.) With $\omega_1 = 1$, $\omega_2 = 0.3 + 1.1i$: for random z_1, z_2 and $z_3 = -z_1 - z_2 + \omega_1 + \omega_2$ the three images have det below 10^{-6} (relative); moving z_3 by 0.13 gives det above $3 \cdot 10^{-3}$.

Exercises

E1 Use T2 to write $\wp(2z)$ in terms of $\wp(z)$ and $\wp'(z)$.

E2 Why are the points with $nP = O$ over $\mathbb{R}$ fewer than n^2?

E3 Show from T1 that $\phi(z)$ is a point where the tangent meets C three times exactly when $3z \in \Lambda$.

Chapter III.3. Counting points over a clock

cubics over $\mathbb{Z}/p \to$ their sizes · from: Chapters II.2, II.4, III.1; Building 0, Chapters 3, 17 · count

Over $\mathbb{Z}/p$ the move system of Chapter III.1 is finite. Its size is $p+1$ plus a sum of p signs, and the signs nearly cancel.

Given

G1 (Chapter III.1.) $C(\mathbb{Z}/p)$ with $\oplus$, for $p \neq 2, 3$ and $\Delta \neq 0$ in $\mathbb{Z}/p$.

G2 (Chapter II.2, T1, T2.) Lagrange; the order of a move divides the size.

G3 (Chapter II.4, T2.) A finite comm move system is $\mathbb{Z}/e_1 \times \cdots \times \mathbb{Z}/e_r$, $e_1 \mid \cdots \mid e_r$, in one way.

G4 (Building 0, Chapter 3, T4, T5.) $\mathbb{Z}/p$ is a field; $c^p = c$.

Definitions

D1 (The square sign.) For c in $\mathbb{Z}/p$: $\chi(c) := 0$ if $c = 0$; 1 if $c = y^2$ for some y; -1 otherwise. $N_p := |C(\mathbb{Z}/p)|$, O included.

Propositions

P1 (How many square roots.) $y^2 = c$ has $1 + \chi(c)$ answers in $\mathbb{Z}/p$. (y and $-y$ differ unless $y = 0$, as $2 \neq 0$; at most two by Building 0, Chapter 7, T4 over $\mathbb{Z}/p$.)

P2 (The count.)

$$N_p \;=\; p + 1 + \sum_{x \in \mathbb{Z}/p} \chi\big(x^3 + ax + b\big).$$

(P1 for each x, and O.)

P3 (Euler's test.) $\chi(c) = c^{(p-1)/2}$ in $\mathbb{Z}/p$, read as 0, 1, -1. (The squares $c \neq 0$ are $(p-1)/2$ in number (P1) and each has $c^{(p-1)/2} = y^{p-1} = 1$ (G4); $x^{(p-1)/2} - 1$ has no more roots than that.)

Theorems

T1 (Lagrange on the curve.) For every P in $C(\mathbb{Z}/p)$, $N_p\, P = O$. (G2.)

T2 (At most two clocks.) $C(\mathbb{Z}/p)$ matches $\mathbb{Z}/e_1 \times \mathbb{Z}/e_2$ with $e_1 \mid e_2$ (G3; at most two factors because the P with $\ell P = O$, ℓ prime, number at most ℓ^2: classical for $\mathbb{Z}/p$; over $\mathbb{C}$ it is Chapter III.2, T4). Also $e_1 \mid p - 1$ (classical, reported).

T3 (The signs nearly cancel. Reported; its road is Building 2.)

$$|N_p - p - 1| \;\leq\; 2\sqrt{p}.$$

The sum in P2 has p terms ± 1 and 0; if they were independent coin tosses its size would be about $\sqrt{p}$ (Building 0, Chapter 17's scale). T3 says the curve's signs never do worse than twice that.

The road: the counts N_{p^m} over the floors with p^m members form a ladder; one rough bound $|N_{p^m} - p^m - 1| \le c\, p^{m/2}$, the same c on every rung, is amplified along the ladder to the constant 2 (Amplify, Building 2). Two inputs of that road are classical and reported: the shape $N_{p^m} = p^m + 1 - \alpha^m - \beta^m$ with $\alpha\beta = p$, and the rough bound.

Remarks

R1 (What the method did.) Counting, and two lines from Part II. The real content, T3, is a cancellation line: the edge the survey of the book keeps meeting (Building 0, Chapter 17, P6).

R2 (Old names.) χ: the Legendre symbol; P3: Euler's criterion; T3: Hasse's theorem (the Riemann hypothesis for elliptic curves over finite fields); α, β: Frobenius eigenvalues; $p + 1 - N_p$: the trace of Frobenius.

R3 (Machine check, `checks/b1-p3-check.py`.) For every prime $5 \le p < 60$ and every smooth (a, b): P2 against direct listing; T1 (each point's order divides N_p); T2 ($e_1 e_2 = N_p$, $e_1 \mid e_2$, $e_1 \mid p - 1$); T3, with the worst case $|N_p - p - 1|/2\sqrt{p} = 0.991$.

Exercises

E1 Count $C(\mathbb{Z}/7)$ for $y^2 = x^3 + 2$ by P2, and check T3.

E2 Show that for $y^2 = x^3 + x$ and $p \equiv 3 \pmod 4$, $N_p = p + 1$. (Pair x with $-x$, and use $\chi(-1) = -1$.)

E3 Use T1 to find the order of $(0, 1)$ on $y^2 = x^3 + 1$ over $\mathbb{Z}/5$ without adding.